A Continuously-Learning Embodied Neural Substrate for Synthetic Affective Agents

Architecture, Mechanisms, and Expected Behavior

Contents

A Continuously-Learning Embodied Neural Substrate for Synthetic Affective Agents: Architecture, Mechanisms, and Expected Behavior

Technical Architecture Whitepaper

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Abstract

We propose a continuously-learning neural substrate architecture for synthetic affective agents, targeting developers building long-running AI systems that must exhibit biological-style emotional continuity, memory consolidation, and self-organization over timescales of months to years. The architecture centers on a novel three-compartment spiking neuron — the **Embodied Multi-Compartment Neuron (EMCN)** — that integrates Izhikevich-style somatic dynamics with dendritic compartmentalization, stochastic ion channel gating, a per-neuron metabolic budget, and a receptor expression vector coupling the neuron to a diffusing three-dimensional chemistry field. Neurons communicate through conductance-based stochastic synapses with per-synapse delays and vesicle pool dynamics. A 3D neuromodulator field replaces scalar hormone variables; emotional state is computed as a readout of this field rather than stored as primary data. Plasticity combines spike-timing-dependent plasticity with chemistry-gated three-factor learning, BCM metaplasticity, and sleep-gated structural plasticity that supports unbounded neurogenesis and apoptosis subject only to hardware constraints. An interoceptive body-state loop provides the embodied signal that drives the chemistry field, completing a substrate–chemistry–body loop that produces emotion as an emergent readout rather than an assigned variable. We describe the full architecture, specify the numerical dynamics, outline a Python-first implementation strategy suitable for a single consumer-class workstation with unified memory (~128 GB RAM, heterogeneous CPU/iGPU/NPU compute), and present expected performance characteristics derived from scaling arguments and component-level benchmarks from the computational neuroscience literature. Expected scale on target hardware is 150,000 to 500,000 EMCN neurons with 10–50 million synapses operating at a 60 Hz external contract with a 1 kHz internal spike loop. We do not present empirical results; the architecture is proposed, and this paper serves as the design specification and projected performance analysis prior to implementation.

Keywords: spiking neural networks, neuromodulation, continuous learning, structural plasticity, synaptic plasticity, affective computing, embodied cognition, biologically plausible learning, computational neuroscience

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1. Introduction

1.1 Motivation

The gap between biologically plausible neural simulations and deployable artificial agents remains wide. Computational neuroscience produces models of striking fidelity — multi-compartment Hodgkin-Huxley neurons embedded in detailed cortical microcircuits, parameterized against electrophysiological recordings — but these models are not embedded in agents that act on the world. Machine learning produces agents of impressive capability — large language models, reinforcement learning systems — but these agents are static after deployment, do not learn from their interactions at the substrate level, and do not have internal states that correspond in any mechanical sense to the emotional and motivational dynamics they often appear to exhibit.

Between these extremes lies an underexplored design space: neural substrates rich enough to support biological dynamics (spike timing, dendritic integration, neuromodulation, structural plasticity, sleep-dependent consolidation) but deployed as the operational substrate of a long-running agent that interacts with an environment over months or years. This paper proposes an architecture for that space.

The motivating use case is a synthetic affective agent — a persistent artificial being that interacts with humans over extended timescales, exhibits emotional and motivational states, remembers individuals, learns from experience, sleeps, and changes over time. Such agents exist today primarily as character simulations driven by large language models with session-level memory and prompt-engineered personas. We argue that the character simulation approach has inherent limitations that a substrate-level approach overcomes: the substrate encodes state, history, and adaptation in the same structures that produce behavior, rather than relying on a retrieval step to inject history into a stateless inference.

The architecture we propose is not a transformer, an LSTM, or a standard spiking neural network. It is closer to a miniature cortical simulation coupled to an interoceptive body model, augmented with the affordances necessary to operate as a real-time agent: snapshot persistence, telemetry for legibility, configurable research modules, and graceful degradation under resource pressure.

1.2 Design goals

The architecture is shaped by six goals, each constraining the design in specific ways:

G1. Biological mechanism over biological metaphor. Existing systems often use biological vocabulary (neurons, hormones, synapses) as labels for algorithmically simple mechanisms (activation functions, scalar variables, weight matrices). We require that the mechanism match the name: a "hormone" must have production, diffusion, receptor binding, and reuptake dynamics; a "synapse" must have conductance kinetics, release probability, and delay; a "neuron" must have compartmentalized dynamics with real integration properties.

G2. Continuous learning as default. Plasticity runs every tick, gated by chemistry rather than by a "training mode" flag. The agent learns whenever it is running. There is no train-then-deploy boundary.

G3. Multi-timescale dynamics. Substrate processes span five orders of magnitude in time: spike dynamics at milliseconds, neuromodulator diffusion at seconds, homeostatic regulation at minutes, sleep-dependent consolidation at hours, and structural change over days and weeks. All are native to the substrate, not bolted on.

G4. Embodiment as emotion substrate. Emotion is the readout of a body-state-driven chemistry field, not an assigned label. A simulated body state closes the loop: chemistry affects the body, body state feeds back into chemistry, and the emotional state of the agent emerges from this loop rather than being set externally.

G5. Unbounded growth with graceful resource scaling. The substrate can add and remove neurons and synapses at any time during operation. The only cap is the hardware envelope, approached through metabolic competition between neurons rather than through hard software limits. This produces a brain that is finite but not fixed.

G6. Deployable as a real-time agent. A 60 Hz external output contract is maintained. The substrate runs on a single consumer-class workstation. Snapshot, rollback, and branching are first-class operations. Telemetry makes internal state legible to human operators at every level.

1.3 What this paper is and is not

This is a **proposal paper**. The architecture has been designed in detail; it has not yet been implemented at full scale. We present expected performance characteristics derived from component-level benchmarks, scaling arguments, and the behavior of related systems in the literature. Empirical validation is future work.

We claim novelty in the **integration** rather than in most individual components. Multi-compartment spiking neurons, chemistry-field neuromodulation, three-factor plasticity, and sleep-gated consolidation are all represented in the literature. What is novel is their unified design inside a single substrate that targets deployment as a real-time embodied agent on consumer hardware, along with several specific mechanisms: the EMCN neuron model as a unified package, the chemistry-as-primary / PAD-as-readout framing, unbounded structural plasticity with metabolic competition as the scale regulator, and a five-tier snapshot scheme for the resulting continuously-changing substrate.

We do not claim that this substrate produces conscious experience, sentience, or understanding. It produces continuous self-modifying neural dynamics coupled to a simulated body, in a configuration that we argue enables behaviors that are harder to produce with other architectures. Whether those behaviors feel lifelike to human observers is an empirical question we expect to investigate, not a claim we make in advance.

1.4 Paper structure

Section 2 surveys related work in spiking neural networks, neuromodulation, continuous learning, and embodied affective architectures. Section 3 gives the architecture overview. Sections 4–10 specify each major component: the EMCN neuron model (4), synapses and topology (5), the chemistry substrate (6), plasticity mechanisms (7), structural plasticity and unbounded growth (8), sleep and consolidation (9), and interoception and embodiment (10). Section 11 covers developmental bootstrapping — how a mature substrate is grown from a compact seed specification. Section 12 specifies the operator telemetry and legibility architecture. Section 13 discusses implementation on a consumer workstation with heterogeneous compute (CPU, iGPU, NPU). Section 14 presents expected performance characteristics. Section 15 lists limitations and open questions. Section 16 concludes.

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2. Related Work

We position this architecture against five bodies of work: spiking neural network simulators, multi-compartment neuron models, neuromodulation and three-factor learning rules, structural plasticity, and embodied affective architectures. We note where we draw on prior work and where we depart from it.

2.1 Spiking neural network simulators

Large-scale SNN simulation has a mature infrastructure. **Brian2** (Stimberg, Brette, and Goodman, 2019) provides a flexible Python framework supporting arbitrary neuron equations and is widely used in computational neuroscience. **NEST** (Gewaltig and Diesmann, 2007) targets very large networks and distributed simulation. **NEURON** (Hines and Carnevale, 1997) is the reference implementation for detailed multi-compartment biophysical simulation. **Arbor** (Akar et al., 2019) is a modern distributed-compute cable simulator. In the machine-learning-adjacent space, **snnTorch** (Eshraghian et al., 2023), **Norse** (Pehle and Pedersen, 2021), and **BindsNET** (Hazan et al., 2018) integrate spiking dynamics with PyTorch for gradient-based training.

Our architecture does not compete with these as general-purpose simulators. It specializes: the substrate is designed for a specific use case (a continuously-running embodied agent on consumer hardware) and trades generality for integration. Where general simulators provide a flexible equation framework and leave the system design to the user, our substrate provides a designed system and exposes configuration rather than equation editing. We expect users who want a full-fidelity multi-compartment Hodgkin-Huxley simulation to use NEURON or Arbor; users who want to train a feedforward SNN to classify MNIST to use snnTorch; and users who want a persistent, embodied, emotionally-modulated agent to use our substrate.

2.2 Multi-compartment neuron models

Point neuron models — leaky integrate-and-fire (LIF), adaptive LIF, and Izhikevich (Izhikevich, 2003) — trade biological detail for computational efficiency. They capture spike timing and basic firing patterns but not dendritic integration. At the opposite extreme, full multi-compartment Hodgkin-Huxley models (Rall, 1959; Hodgkin and Huxley, 1952) capture dendritic computation but are computationally expensive: a single cortical pyramidal neuron simulated with realistic morphology can require milliseconds of compute per simulated millisecond of neural activity on modern CPUs.

Reduced multi-compartment models occupy the middle ground. The two-compartment Pinsky-Rinzel model (Pinsky and Rinzel, 1994) captures key interactions between soma and apical dendrite at a fraction of the cost of full cable models. The Larkum, Zhu, and Sakmann (1999) work established that NMDA-dependent apical dendritic plateaus play a critical role in cortical computation, motivating the inclusion of NMDA plateau dynamics in reduced models. More recently, Beniaguev, Segev, and London (2021) demonstrated that a single cortical pyramidal neuron can perform computations equivalent to a 5-8 layer deep neural network, arguing for the importance of dendritic nonlinearity in any biologically grounded model.

The EMCN model we propose occupies this reduced-multi-compartment middle ground. It is closer to a three-compartment Pinsky-Rinzel variant with Izhikevich soma dynamics than to either pure Izhikevich or full HH. We add stochastic channel gating (following the tradition of Goldwyn and Shea-Brown, 2011) to produce realistic spike timing variability, and a metabolic budget that has no direct analog in standard neuron models but captures the well-established relationship between neural activity and metabolic cost (Attwell and Laughlin, 2001).

2.3 Neuromodulation and three-factor learning rules

Standard Hebbian and spike-timing-dependent plasticity (STDP; Bi and Poo, 1998; Song, Miller, and Abbott, 2000) describe how synaptic weights change as a function of pre- and post-synaptic activity. A substantial body of literature extends this to include a third factor representing neuromodulatory state or reward: dopamine-modulated STDP (Izhikevich, 2007; Frémaux and Gerstner, 2016), eligibility-trace approaches (Sutton and Barto, 1998; Gerstner et al., 2018), and explicit three-factor rules (Kuśmierz, Isomura, and Toyoizumi, 2017).

More biologically detailed work addresses the spatiotemporal dynamics of neuromodulators. Arbuthnott and Wickens (2007) describe the volume transmission of dopamine through the striatum. Fuxe et al. (2010) review the broader evidence for extra-synaptic neuromodulator dynamics. The computational consequences of this spatial structure are less well-explored but include the suggestion that local chemistry gradients could implement credit assignment in ways simple scalar modulation cannot.

Our chemistry substrate takes this spatial view seriously. Rather than treating dopamine as a scalar variable modulating plasticity globally, we model it as a 3D concentration field with diffusion, uptake, and receptor binding dynamics. The three-factor plasticity rule then reads local chemistry at the synapse, not a global modulator variable. This is closer to biology and, we argue, produces different network-level behaviors: learning is spatially heterogeneous in ways scalar-modulator approaches cannot capture.

2.4 Structural plasticity

Most SNN work treats network topology as fixed. Biological networks are not fixed: real brains add and remove synapses continuously, add new neurons in specific regions throughout life (Eriksson et al., 1998; Spalding et al., 2013; Boldrini et al., 2018), and reorganize connectivity over longer timescales.

Computational models of structural plasticity are less developed than those of synaptic plasticity, but foundational work exists. Butz, Wörgötter, and van Ooyen (2009) review computational models of structural plasticity and propose unified frameworks. Fauth, Wörgötter, and Tetzlaff (2015) explore the interaction between synaptic and structural plasticity in network function. More recent work (Diaz-Pier et al., 2016) integrates structural plasticity with large-scale simulation in NEST.

A key difference between our architecture and most prior structural plasticity work is that we do not cap the network size. Prior work typically operates within a fixed neuron pool with variable connectivity. We allow the neuron pool itself to grow and shrink, subject only to hardware constraints expressed as metabolic competition. This is a design choice motivated by the target use case (long-running agents that should be able to develop over time) and requires a specific implementation strategy (arena-based slotmaps with generational indexing) that we describe in Section 8 and Section 13.

2.5 Embodied affective architectures

The affective computing literature (Picard, 1997; Calvo, D'Mello, Gratch, and Kappas, 2015) spans a wide range of architectures from rule-based to statistical to neural. Of particular relevance to our work:

- **PAD-space emotion models** (Mehrabian, 1996; Russell, 2003) provide a dimensional (pleasure, arousal, dominance) framework that we adopt as an output representation.
- **Appraisal theories** (Scherer, 2009; Ortony, Clore, and Collins, 1988) describe how emotional responses arise from cognitive evaluation of events. Our architecture is compatible with appraisal models at the cognitive layer, which sits above the substrate.
- **Interoceptive inference** (Seth, 2013; Barrett, 2017) frames emotion as the brain's inference about body state. This tradition directly motivates our design: the chemistry field is driven in part by simulated body state, and emotion is read out from chemistry.
- **Neuroscientifically grounded embodied agents** in the tradition of Damasio (1994, 1999) argue that emotion cannot be separated from body. Our interoception module implements this view as a concrete computational mechanism.

We are not aware of prior architectures that combine biologically-detailed spiking dynamics, spatial chemistry modulation, continuous structural plasticity, sleep consolidation, and interoceptive embodiment in a single deployable system. The individual ingredients are established; the integration is, to our knowledge, new.

2.6 Sleep and memory consolidation

The hypothesis that sleep plays an active role in memory consolidation is well-supported (Diekelmann and Born, 2010; Rasch and Born, 2013). Specific mechanisms include sharp-wave ripples in hippocampus during NREM sleep (Buzsáki, 2015), replay of waking activity patterns (Wilson and McNaughton, 1994), and synaptic homeostasis via downscaling (the SHY hypothesis; Tononi and Cirelli, 2014).

Our sleep substrate implements these mechanisms not as scripted events but as consequences of a chemistry regime shift. During deep NREM, low acetylcholine combined with sharp-wave ripple-like activity in hippocampal-analog regions produces selective plasticity that preferentially consolidates recent experiences. During REM, high acetylcholine combined with low aminergic tone produces dreamlike replay with active plasticity. We describe this in detail in Section 9.

2.7 Position

This work sits at the intersection of computational neuroscience (which emphasizes biological fidelity), embodied cognitive architecture (which emphasizes the body-brain loop), and applied ML systems engineering (which emphasizes deployability). It is not purely any of these. We prioritize biological mechanism where mechanism shapes behavior in ways that algorithmic approximations miss, we prioritize deployment on consumer hardware as a first-class constraint, and we accept the loss of generality that specialization entails.

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3. Architecture Overview

3.1 Four layers

The system comprises four layers, each operating at different timescales and on different compute resources.

Presentation layer. External contract. Produces a 60 Hz output stream containing PAD readout, dominant emotion label, decision signals, expression parameters, and audio. Receives sensory input from external sources. This layer implements APIs, avatar rigs, text-to-speech, and whatever external integrations a specific deployment requires. It is not substrate; it is downstream of substrate.

Cognitive layer. Higher-level structured reasoning: planning, language formatting via a large language model acting as an output device (not an input to the emotional loop), memory indexing and retrieval, social bond tracking, autonomous behavior selection. This layer runs at event and frame timescales on the CPU, in Python. It queries the substrate for state and sends events to the substrate, but does not directly drive substrate dynamics.

Substrate layer. The core neural system. Runs the EMCN neurons, conductance synapses, chemistry field, glial cells, plasticity rules, sleep state machine, interoception simulator. Operates internally at a 1 kHz spike loop, a 10 Hz chemistry loop, and slower homeostatic and structural loops. Exposes state via queries and accepts external events. This is where the majority of compute budget is spent.

Perception layer. Sensory preprocessing: speech-to-text, face analysis, gaze tracking, prosody extraction, vision. Runs continuously on sampled input streams. Feeds features into the substrate via dedicated input pathways. Operates at event rates dictated by external sensors. Uses the NPU on supported hardware for dedicated ML inference.

The substrate layer is the novel contribution. The other three layers exist in many systems and are included here for completeness.

3.2 Timescale hierarchy

The substrate runs processes at five distinct timescales:

Process	Rate	Notes
Spike integration	1 kHz (or sub-stepped from 60 Hz)	Neuron voltage and channel updates
Frame readout	60 Hz	External contract, PAD and emotion readout
Chemistry diffusion	10 Hz	3D voxel field updates
Homeostatic regulation	0.1 Hz	BCM thresholds, glial scaling, criticality control
Sleep-gated structural	1 per sleep cycle	Neurogenesis, apoptosis, synaptogenesis, pruning

Each process has a separate scheduling mechanism and compute budget. The 60 Hz contract is the external commitment; the 1 kHz spike loop is internal and not visible outside the substrate. The chemistry and homeostatic loops run in background threads or on separate compute resources (GPU for chemistry diffusion, CPU for homeostatic bookkeeping). Structural changes happen in batches during sleep phase transitions, when the substrate is in a well-defined quiescent state.

3.3 Hardware model

Target hardware is a single-node consumer workstation with three compute domains sharing unified memory:

- **CPU**: 16-core general-purpose processor. Runs the 60 Hz orchestration loop, cognitive layer, event queue, and chemistry uptake bookkeeping. Allocates specific cores to specific roles.
- **iGPU**: Integrated GPU with ~40 compute units and direct access to system RAM. Runs substrate hot paths (neuron integration, synapse propagation, chemistry diffusion, STDP updates). Also time-shares for large language model inference (on the cognitive layer path) and text-to-speech synthesis.
- **NPU**: Neural processing unit optimized for standard ML inference (~50 TOPS). Runs perception layer: speech-to-text, face and gaze analysis, prosody, embedding models for semantic memory indexing.

Total system memory is approximately 128 GB, shared across all compute domains. The unified memory architecture is load-bearing for performance: large neural state structures can be accessed by both CPU and iGPU without explicit copies, which is the difference between the substrate being feasible at 60 Hz with 500 thousand neurons and not being feasible.

3.4 Substrate topology

The substrate occupies a three-dimensional volume of approximately 10 mm × 10 mm × 5 mm in simulated physical space. Neurons have positions within this volume. Chemistry is a voxel grid (128 × 128 × 128, giving ~2.1 million voxels) covering the same volume. Synapses connect neurons across the volume, with conduction delays scaled by separation distance. Glial cells are sparsely distributed through the volume.

The volume is loosely zoned into functional regions (amygdala-analog, hippocampus-analog, prefrontal-analog, sensory-integration-analog, and several generic cortex-analog regions), but these zones are not anatomically rigid: they bias initial neuron placement and early connectivity growth. Functional specialization emerges from the interaction between embodied input, chemistry dynamics, and activity-dependent plasticity during a developmental phase, not from a hand-designed circuit.

The substrate does not correspond to any specific real brain region or organism. It is designed to support the dynamics we need, not to reproduce biology.

3.5 State variables

The substrate's full state comprises the following structures:

Per neuron (EMCN state, detailed in Section 4): - Position in 3D volume - Cell type tag (from a taxonomy of ~12 types) - Soma compartment state: voltage, recovery variable, calcium concentration, ATP level, adaptive threshold, time of last spike - Apical dendrite compartment state: voltage, NMDA plateau state, calcium - Basal dendrite compartment state: voltage, calcium - Channel gating states (when stochastic channels active) - Receptor expression vector (one entry per chemical) - Generational ID (for slotmap reuse after apoptosis)

Per synapse: - Pre- and post-synaptic neuron IDs, post-synaptic compartment - Maximum conductance, reversal potential, rise and decay time constants - Conduction delay (in spike-loop ticks) - Release probability, vesicle pool size, replenishment rate - STDP pre- and post-traces, eligibility trace - Spine size (structural proxy) - Generational ID

Per chemistry voxel: - Concentration vector (one entry per chemical, typically 8 base chemicals or 14 with reproductive module enabled)

Per glial cell: - Position - Metabolic support capacity - Scaling and pruning flags

Interoceptive body state: - Heart rate, heart rate variability, respiration rate and phase, gut activity, core temperature, muscle tension, interoceptive noise

Global state: - Substrate age (ticks since creation) - Sleep phase state machine state - Criticality controller state - Developmental phase indicators

Total hot state for a 250,000-neuron substrate with 20 million synapses and full chemistry is approximately 1.5–2 GB. Working memory for plasticity traces, propagation buffers, and telemetry roughly doubles this.

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4. The EMCN Neuron Model

This section specifies the **Embodied Multi-Compartment Neuron (EMCN)** in full. The EMCN is the unit neuron of the substrate. All substrate neurons are EMCNs; different cell types are EMCNs with different parameter sets.

4.1 Design philosophy

The EMCN is designed to be the minimum neuron model that supports the full range of substrate dynamics we require. We add complexity only where it enables behavior that a simpler model cannot produce:

- **Three compartments** (soma, apical dendrite, basal dendrite) because dendritic integration shapes computation (Larkum, Zhu, and Sakmann, 1999) and a single compartment cannot represent NMDA plateau dynamics or compartmental plasticity differences.
- **Izhikevich-family soma dynamics** because they produce the full range of cortical cell types at a fraction of Hodgkin-Huxley cost (Izhikevich, 2003, 2004).
- **Stochastic channel gating option** because deterministic neurons cannot reproduce biological spike timing variability (Goldwyn and Shea-Brown, 2011).
- **Metabolic budget** because energy constraints shape biological neural dynamics (Attwell and Laughlin, 2001) and we need a principled source of fatigue and a mechanism coupling neurons to glial support.
- **Receptor expression vector** because this is the mechanism through which a neuron couples to the chemistry field, which is central to the architecture.

We do not include: full cable dynamics (too expensive for the scale we target), explicit synaptic spine morphology (the synapse carries a spine size proxy), or mitochondrial dynamics (abstracted into ATP).

4.2 Compartment dynamics

Each EMCN has three compartments connected by axial conductances. The soma implements Izhikevich-family dynamics; the dendrites are passive-plus-NMDA-plateau.

4.2.1 Soma The soma voltage v_s and recovery variable u_s follow the Izhikevich (2003) form:

$$\frac{dv_s}{dt} = 0.04v_s^2 + 5v_s + 140 - u_s + I_{\text{syn,s}} + I_{\text{ax,a}} + I_{\text{ax,b}} + I_{\text{intero}} + \xi_s(t)$$

$$\frac{du_s}{dt} = a(bv_s - u_s)$$

with spike reset:

$$\text{if } v_s \geq v_{\text{peak}}: \quad v_s \leftarrow c, \quad u_s \leftarrow u_s + d$$

where: - $I_{\text{syn,s}}$ is the synaptic current onto the soma (from inhibitory synapses, which preferentially target soma) - $I_{\text{ax,a}}$ and $I_{\text{ax,b}}$ are axial currents from the apical and basal dendrites - I_{intero} is the interoceptive current (nonzero only for interoception-integrating neurons) - $\xi_s(t)$ is a noise term, either from explicit stochastic channel gating (Section 4.5) or Langevin background noise - (a, b, c, d) are cell-type parameters from the Izhikevich taxonomy - v_{peak} is the spike threshold (typically 30 mV)

The axial currents couple the compartments:

$$I_{\text{ax,a}} = g_{\text{ax,a}}(v_a - v_s), \quad I_{\text{ax,b}} = g_{\text{ax,b}}(v_b - v_s)$$

with $g_{\text{ax,a}} \approx 0.5$ and $g_{\text{ax,b}} \approx 0.8$ in the reference parameterization.

4.2.2 Apical dendrite The apical dendrite carries most excitatory input from distal sources. Its dynamics include the NMDA plateau, which is the defining feature of pyramidal-neuron dendritic computation:

$$\tau_a \frac{dv_a}{dt} = -(v_a - v_{\text{rest}}) + g_{\text{ax,a}}(v_s - v_a) + I_{\text{syn,a}} + I_{\text{NMDA}}(v_a, [\text{Ca}^{2+}]_a)$$

The NMDA plateau current is a nonlinear function of apical voltage and local calcium, producing a prolonged depolarization when both exceed threshold:

$$I_{\text{NMDA}} = g_{\text{NMDA}} \cdot \sigma(v_a, V_{\text{plat}}) \cdot \sigma([\text{Ca}^{2+}]_a, \text{Ca}_{\text{plat}})$$

where σ is a sigmoid function and the plateau thresholds V_{plat} and Ca_{plat} are set such that the plateau requires coincident depolarization and calcium elevation. The plateau lasts approximately 50–100 ms once triggered, during which it substantially elevates somatic firing probability.

The plateau is what gives apical dendrites coincidence-detection behavior: inputs that arrive simultaneously produce disproportionately larger somatic responses than inputs that arrive separately. This is central to our dendritic compartmentalization and has direct consequences for the plasticity mechanism (Section 7).

4.2.3 Basal dendrite The basal dendrite is simpler, with passive cable-like dynamics:

$$\tau_b \frac{dv_b}{dt} = -(v_b - v_{\text{rest}}) + g_{\text{ax,b}}(v_s - v_b) + I_{\text{syn,b}}$$

No NMDA plateau. Basal dendrites carry the majority of routine excitatory input and provide graded depolarization to the soma without the coincidence-detection gating of the apical compartment.

4.3 Calcium dynamics

Calcium is tracked separately in each compartment and drives plasticity. Calcium enters during spikes (soma) and during synaptic NMDA activation (dendrites), and is extruded exponentially:

$$\frac{d[\text{Ca}^{2+}]_c}{dt} = -\frac{[\text{Ca}^{2+}]_c}{\tau_{\text{Ca}}} + J_{\text{Ca,c}}(t)$$

where $c \in \{s, a, b\}$ indexes compartment and $J_{\text{Ca,c}}$ is the compartment-specific calcium influx. Typical calcium decay time constant $\tau_{\text{Ca}} \approx 100$ ms, with compartment-specific values.

Calcium concentration is the signal that converts transient electrical activity into durable synaptic change. High-frequency or coincident inputs produce high calcium; low-frequency or isolated inputs do not. The three-factor plasticity rule (Section 7) reads calcium, not voltage or spike count directly.

4.4 Metabolic budget (ATP)

Each EMCN carries a scalar ATP level $\alpha \in [0, 1]$ representing metabolic availability:

$$\frac{d\alpha}{dt} = -k_{\text{base}} - k_{\text{spike}} \cdot \delta_{\text{spike}}(t) + k_{\text{supply}} \cdot g(\mathbf{x})$$

where: - k_{base} is the basal metabolic rate (always draining) - k_{spike} is the per-spike metabolic cost - $\delta_{\text{spike}}(t)$ is a spike indicator - k_{supply} is the supply rate - $g(\mathbf{x})$ is the local glial support density at the neuron's position

ATP is bounded to $[0, 1]$. Behavior when ATP is low:

- At $\alpha < 0.3$: adaptive threshold elevated (neuron requires stronger input to spike)
- At $\alpha < 0.1$: plasticity gate closed (no LTP or LTD)
- At $\alpha < 0.05$: candidate for apoptosis

ATP is the substrate's native representation of fatigue. A neuron that has been firing hard has low ATP, fires less reliably, learns less, and is at elevated apoptosis risk. Sleep (via increased glial supply and reduced activity) replenishes ATP.

The global distribution of ATP across neurons is a finite resource that couples to glial density. This is the mechanism through which we implement competition: when the substrate approaches its metabolic ceiling, ATP supply per neuron drops, and low-contribution neurons lose the competition for resources and are pruned (Section 8).

4.5 Stochastic channel gating (optional per cell type)

For cell types where spike timing variability is important — notably fast-spiking interneurons — we support explicit stochastic ion channel gating. Each relevant channel type has a Markov chain representation:

$$P(\text{open}) \to P(\text{closed}) \text{ with rate } \alpha_{\text{open}}(v), \quad P(\text{closed}) \to P(\text{open}) \text{ with rate } \beta_{\text{open}}(v)$$

With a finite number of channels N_{ch} per compartment (typically 50–200), the total conductance at each timestep is a binomial draw from the channel population:

$$g_{\text{ch}}(t) = g_{\max} \cdot \frac{N_{\text{open}}(t)}{N_{\text{ch}}}$$

This produces realistic spike timing jitter without an ad-hoc noise term. For neurons where this detail is not needed, a Langevin noise term approximates the effect at much lower cost.

The choice between stochastic channels and Langevin noise is made per cell type in configuration. We expect the default substrate to use stochastic channels in fast-spiking interneurons and Langevin noise elsewhere.

4.6 Chemistry receptor coupling

Each EMCN has a receptor expression vector $\mathbf{r} \in \mathbb{R}^{N_{\text{chem}}}$ where N_{chem} is the number of chemicals (8 or 14 depending on configuration). Each component r_k represents the sensitivity of this neuron to chemical k.

At each neuron update, the neuron samples local chemistry at its position via trilinear interpolation from the voxel grid:

$$\mathbf{c}_{\text{local}} = \text{TrilinearInterp}(\mathbf{C}, \mathbf{x}_{\text{neuron}})$$

where $\mathbf{C}$ is the voxel chemistry tensor. The receptor effect is the element-wise product:

$$\eta = \mathbf{r} \odot \mathbf{c}_{\text{local}}$$

This effect vector η modulates several neuron parameters:

- **Gain**: $G_{\text{eff}} = 1 + \mathbf{w}_G^T \eta$ multiplies synaptic input currents
- **Threshold**: $v_{\text{peak,eff}} = v_{\text{peak}} + \mathbf{w}_\theta^T \eta$
- **Plasticity rate**: $\eta_{\text{plast,eff}} = \eta_{\text{plast}} \cdot (1 + \mathbf{w}_P^T \eta)$

The weight vectors $\mathbf{w}_G, \mathbf{w}_\theta, \mathbf{w}_P$ are set by cell type and are typically small (modulations are at the tens-of-percent scale, not order-of-magnitude). The net effect is that a neuron in high local dopamine has elevated gain and plasticity; a neuron in high cortisol has shifted threshold and altered plasticity.

This receptor coupling is the primary mechanism by which the chemistry field exerts its effect on the substrate. It is also the mechanism that enables the receptor expression vector to be a learnable feature — cells in specific regions can develop receptor profiles matching the chemistry they encounter, a form of homeostatic receptor regulation that we defer to future work.

4.7 Cell type taxonomy

The substrate uses a taxonomy of approximately 12 cell types, each characterized by a parameter set. The taxonomy draws on the Izhikevich (2004) catalog and augments with EMCN-specific parameters:

Type	Role	Izhikevich params	Stochastic channels	Compartment emphasis
Regular spiking pyramidal (RS)	General cortical excitation	(0.02, 0.2, -65, 8)	No (Langevin)	Apical + basal
Intrinsically bursting (IB)	Deep-layer pyramidal	(0.02, 0.2, -55, 4)	No	Apical + basal
Chattering (CH)	Cortical gamma generation	(0.02, 0.2, -50, 2)	No	Apical + basal
Fast spiking interneuron (FS)	Inhibitory, fast gamma	(0.1, 0.2, -65, 2)	Yes	Soma-dominated
Low-threshold spiking (LTS)	Inhibitory, theta	(0.02, 0.25, -65, 2)	No	Basal
Thalamocortical (TC)	Sensory input relay	(0.02, 0.25, -65, 0.05)	No	Basal
Resonator (RZ)	Oscillator cells	(0.1, 0.26, -65, 2)	No	Apical
Basket cell	Fast inhibitory, perisomatic	(0.1, 0.2, -65, 4)	Yes	Soma-dominated
Martinotti cell	Inhibitory, ascending	(0.02, 0.25, -60, 2)	No	Apical-inhibitory
VIP interneuron	Disinhibitory	(0.02, 0.3, -65, 4)	No	Basal
Granule cell (DG-analog)	Hippocampal pattern separation	(0.01, 0.3, -65, 8)	No	Small apical
Pyramidal CA3-analog	Hippocampal pattern completion	(0.02, 0.2, -65, 8)	No	Apical recurrent

Each type also has type-specific values for: compartment membrane time constants, NMDA plateau thresholds, calcium dynamics, ATP cost, base firing rate target, receptor expression baseline, and growth/pruning susceptibility.

The taxonomy is not sacred; it is a working set. Cell types can be added, modified, or pruned based on observed substrate behavior. The design intent is to give the substrate a sufficiently heterogeneous neuron population that rich dynamics can emerge, without over-committing to a specific biological ground truth.

4.8 Integration

EMCN dynamics are integrated with a sub-stepped explicit scheme:

- Outer tick at 60 Hz (16.67 ms)
- Sub-steps at 1 kHz (1 ms) — 16-17 sub-steps per outer tick
- Within each sub-step, compartment voltages updated with forward Euler
- Spike detection after each sub-step
- Calcium and ATP updates at sub-step rate
- Chemistry receptor coupling evaluated once per outer tick (chemistry changes slowly; sampling at 60 Hz is sufficient)
- Stochastic channel updates at sub-step rate for cell types using them

This sub-stepping is essential for stability: EMCN dynamics with Izhikevich soma and NMDA plateau are stiff enough that 16.67 ms steps produce numerical artifacts. A 1 ms sub-step is biologically appropriate (real neurons have sub-millisecond timescales for channel dynamics) and computationally affordable given modern GPU throughput.

4.9 Computational cost per neuron

The per-neuron computational cost per 1 kHz sub-step, on modern GPU hardware with SIMD execution, is estimated from component-level benchmarks:

- Izhikevich soma update: ~15 FLOPs
- Two passive dendrite updates: ~10 FLOPs each
- NMDA plateau evaluation: ~20 FLOPs (includes sigmoid)
- Axial coupling (3 pairs): ~10 FLOPs
- Calcium dynamics (3 compartments): ~15 FLOPs
- ATP update: ~5 FLOPs
- Receptor coupling (amortized to 1/60 sub-steps): ~(16 × N_chem) / 16 = ~100 FLOPs amortized
- Stochastic channels (when enabled): ~50 FLOPs

Total per EMCN per sub-step: approximately 200–300 FLOPs, with stochastic channels adding roughly 50 more.

For 250,000 neurons at 1 kHz: $250{,}000 \times 1000 \times 250 = 6.25 \times 10^{10}$ FLOPs/sec = 62.5 GFLOPs/sec for neuron integration alone. Well within the capability of a single consumer iGPU, but non-trivial; synapse propagation and chemistry diffusion add substantial additional cost (Section 14).

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5. Synapse Model and Network Topology

5.1 Conductance-based synapses

Synapses in the substrate are not scalar weights. Each synapse is a full object with kinetics, delay, and stochastic release dynamics. When a pre-synaptic spike arrives at a synapse, a conductance pulse is injected onto the target compartment with dual-exponential rise and decay:

$$g_{\text{syn}}(t) = g_{\max} \cdot s \cdot \left(e^{-(t-t_{\text{arrival}})/\tau_{\text{decay}}} - e^{-(t-t_{\text{arrival}})/\tau_{\text{rise}}}\right)$$

where s is the spine size factor (structural proxy, see below), t_{arrival} is the post-delay arrival time, and $\tau_{\text{rise}}, \tau_{\text{decay}}$ are synapse-type-specific time constants.

The current into the post-synaptic compartment from this synapse is:

$$I_{\text{syn}}(t) = g_{\text{syn}}(t) \cdot (v_{\text{post}} - E_{\text{rev}})$$

where E_{rev} is the reversal potential. This captures the driving force correctly: current flows toward the reversal potential and vanishes when the post-synaptic voltage equals it. Excitatory synapses have $E_{\text{rev}} \approx 0$ mV (depolarizing); inhibitory synapses have $E_{\text{rev}} \approx -75$ mV (hyperpolarizing).

5.2 Synapse types

We use three primary synapse types corresponding to the dominant fast neurotransmitter receptors in cortex:

Type	E_{rev}	τ_{rise}	τ_{decay}	Notes
AMPA-like (fast excitatory)	0 mV	0.5 ms	5 ms	Dominant onto basal and apical
GABA-A-like (fast inhibitory)	-75 mV	0.5 ms	8 ms	Perisomatic from basket cells
NMDA-like (slow excitatory)	0 mV	5 ms	80 ms	Voltage-dependent Mg block

NMDA synapses have a voltage-dependent magnesium block:

$$g_{\text{NMDA,eff}} = g_{\text{NMDA}} \cdot \frac{1}{1 + ([\text{Mg}]/3.57) \cdot e^{-0.062 v_{\text{post}}}}$$

This is the standard formulation (Jahr and Stevens, 1990) and captures the "coincidence detector" behavior of NMDA receptors: they require both presynaptic release and postsynaptic depolarization to conduct. This is crucial for dendritic plateau dynamics and for calcium-dependent plasticity.

The synapse type is set at creation based on the pre-synaptic neuron's identity (Dale's law: a given neuron releases only one primary transmitter) and the post-synaptic compartment (inhibitory synapses preferentially target soma or proximal dendrite; excitatory synapses target basal or apical dendrite).

5.3 Stochastic release

Synaptic transmission is not deterministic. Each spike arrival triggers a probabilistic release:

$$P(\text{release}) = P_{\text{release}}$$

where $P_{\text{release}} \in [0, 1]$ is the current release probability for this synapse. Typical baseline values are 0.3–0.5, varying with synapse type and short-term history.

If release fails, no conductance pulse is injected. If release succeeds, the vesicle pool is decremented:

$$N_{\text{vesicle}} \leftarrow N_{\text{vesicle}} - 1$$

When the vesicle pool is depleted, further release attempts fail regardless of release probability, capturing short-term synaptic depression. The pool replenishes with a time constant $\tau_{\text{replenish}}$:

$$\frac{dN_{\text{vesicle}}}{dt} = \frac{N_{\max} - N_{\text{vesicle}}}{\tau_{\text{replenish}}}$$

Typical values: $N_{\max} = 5$ readily-releasable vesicles per synapse, $\tau_{\text{replenish}} \approx 500$ ms.

Stochastic release produces several behaviors that deterministic synapses cannot:

- Single-trial variability in post-synaptic responses
- Short-term depression under high-frequency presynaptic firing
- Short-term facilitation (via release probability dynamics we have elided for brevity but include in the full model)
- A natural source of variance that couples into downstream firing patterns

5.4 Per-synapse delays

Each synapse has a transmission delay representing the time for an action potential to travel from the pre-synaptic soma to the synaptic site. Delays are computed at synapse creation based on the distance between pre- and post-synaptic neurons and a conduction velocity parameter:

$$d_{\text{syn}} = \lfloor d_{\text{baseline}} + \frac{\|\mathbf{x}_{\text{pre}} - \mathbf{x}_{\text{post}}\|}{v_{\text{cond}}} \cdot f_{\text{tick}} \rfloor$$

where d_{baseline} is a minimum delay (typically 1 ms), v_{cond} is the conduction velocity (typically 0.3–1.0 m/s for unmyelinated local axons), and f_{tick} is the spike-loop tick frequency (1000 Hz). Typical delays range from 1 to 10 ms, rarely exceeding 20 ms for the longest connections in the 10 mm volume.

Delays are implemented as ring buffers per delay class: synapses with the same integer delay share a queue, and at each tick the queue for delay d is drained into post-synaptic compartments.

5.5 Gap junctions

In addition to chemical synapses, the substrate supports electrical synapses (gap junctions) between specific interneuron populations. Fast-spiking interneurons in particular are extensively coupled by gap junctions in cortex, which is essential for gamma-band oscillations (Galarreta and Hestrin, 1999; Gibson, Beierlein, and Connors, 1999).

Gap junctions are modeled as bidirectional ohmic conductances:

$$I_{\mathrm{gap},i} = g_{\mathrm{gap}}(v_j - v_i)$$

with typical $g_{\mathrm{gap}} \approx 0.1$ nS. Gap junctions are restricted at creation to specific interneuron types, matching biology, and are not subject to STDP (their plasticity, when present, is slower and not well-characterized).

5.6 Topology

Initial connectivity is random with distance-dependent probability:

$$P(\mathrm{connect}(i,j)) = P_0(\mathrm{type}_i, \mathrm{type}_j) \cdot e^{-\|\mathbf{x}_i - \mathbf{x}_j\|/\lambda_{\mathrm{conn}}}$$

where P_0 depends on cell types (following rough cortical connection probability tables from the literature, e.g., Markram et al., 2015) and λ_{conn} is a length constant. Typical result: sparse connectivity with ~0.1–1% of possible pairs connected, ~100–500 synapses per neuron, heavy bias toward local connections.

The substrate enforces Dale's law: each neuron is either excitatory (releases glutamate-analog onto excitatory synapses) or inhibitory (releases GABA-analog). The ratio is approximately 80/20 E/I, matching cortex.

Connectivity is not fixed. After the developmental phase, activity-dependent synaptogenesis (Section 8) grows new connections between co-active neurons, and pruning removes underused connections. The mature substrate has a topology that reflects its history, not its initial design.

5.7 Sparse representation and computational cost

Synapse storage uses a compressed sparse row (CSR) representation indexed by post-synaptic neuron, which matches the access pattern of synapse propagation (for each post-synaptic neuron, iterate over incoming synapses). For structural plasticity, we maintain auxiliary indices to support efficient add and remove.

Per-tick synapse propagation cost, given sparse connectivity:

$$C_{\text{syn}} = N_{\text{spikes per tick}} \cdot \bar{k}_{\text{out}} \cdot C_{\text{per-site}}$$

where $N_{\text{spikes per tick}}$ is the average number of spikes per 1 ms tick (for a 100 kHz population rate across 250 k neurons, approximately 100), $\bar{k}_{\text{out}}$ is the average out-degree (~200), and $C_{\text{per-site}}$ is the cost of processing one spike at one synapse (~20 FLOPs including release stochasticity and conductance injection).

Total: $C_{\text{syn}} \approx 100 \times 200 \times 20 = 4 \times 10^5$ FLOPs per tick, or 4×10^8 FLOPs/sec. Small compared to neuron integration, and well-suited to event-driven sparse processing on GPU.

5.8 Activity-dependent delay adjustment (optional)

Real axonal conduction delays are not fixed; they can be tuned by myelination, which is subject to activity-dependent regulation (Fields, 2015; Gibson et al., 2014). We provide optional activity-dependent delay adjustment as an experimental feature: synapses that carry consistently well-timed spikes (coincident with post-synaptic activity) have their delays slightly decremented; those that carry poorly-timed spikes have delays slightly increased.

This is computationally cheap (a scalar adjustment on a small fraction of synapses per sleep cycle) and captures an aspect of biological plasticity that is usually omitted from computational models. We mark it as optional because its contribution to overall substrate function is not yet characterized and it introduces a new axis of variability that may complicate reproducibility.

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6. Chemistry Substrate

6.1 Why spatial chemistry

Many SNN architectures that model neuromodulation do so via scalar global variables: "dopamine level = 0.7" modulates plasticity globally. This is an abstraction that collapses important biological structure. Real neuromodulators:

- Diffuse through extracellular space with specific diffusion coefficients
- Have finite lifetimes due to reuptake and enzymatic degradation
- Are produced locally by specific cell populations
- Act via receptor subtypes with distinct binding kinetics and downstream effects
- Create spatial gradients that differ across brain regions and across timescales

Collapsing all of this to a scalar hides mechanisms that matter for computation. Credit assignment, for example: a dopamine burst in response to reward does not modulate plasticity uniformly across the whole brain; it modulates plasticity in regions where dopamine actually rises, which is a function of which dopaminergic neurons were active and where they project.

Our chemistry substrate takes the spatial view: we simulate a 3D concentration field for each modulator, with production, diffusion, uptake, and receptor binding. This is expensive compared to scalar modulators but affords behaviors scalar modulators cannot produce, and the expense is manageable on modern hardware.

6.2 Voxel grid

Chemistry is represented on a 128 × 128 × 128 voxel grid covering the substrate volume. Each voxel stores concentration values for N_{chem} chemicals (8 base chemicals, optionally expanded to 14 with the reproductive endocrine module). Total field storage: $128^3 \times 8 \times 4$ bytes ≈ 67 MB for the base configuration.

The voxel size (~80 μm per edge) is larger than a single synapse but smaller than a typical neuron's dendritic arbor, giving the field enough resolution to resolve local gradients around active regions while remaining computationally tractable.

6.3 Chemical inventory

The 8 base chemicals and their primary roles:

Chemical	Primary role	Time constants (production, decay)	Diffusion coefficient
Dopamine	Reward, salience	1 s / 8 s	Medium (200 μm²/s)
Serotonin	Mood, stability	10 s / 30 s	Slow (100 μm²/s)
Cortisol	Stress	60 s / 300 s	Very slow (50 μm²/s)
Oxytocin	Social binding	10 s / 60 s	Medium (150 μm²/s)
Adrenaline	Acute arousal	1 s / 15 s	Fast (300 μm²/s)
GABA-peptide	Calm, inhibition	2 s / 10 s	Fast (300 μm²/s)
Melatonin	Sleep drive	60 s / 600 s	Very slow (40 μm²/s)
Acetylcholine	Attention, plasticity gate	1 s / 5 s	Medium (250 μm²/s)

These represent modulatory neuropeptides and small molecules that act via volume transmission, not the fast synaptic transmitters (glutamate, GABA) which are handled at the synapse level.

6.4 Field dynamics

The concentration of chemical k at voxel $\mathbf{v}$ evolves according to a reaction-diffusion equation:

$$\frac{\partial c_k(\mathbf{v},t)}{\partial t} = D_k \nabla^2 c_k(\mathbf{v},t) - \gamma_k c_k(\mathbf{v},t) - U_k(\mathbf{v}) c_k(\mathbf{v},t) + S_k(\mathbf{v},t)$$

where: - D_k is the diffusion coefficient - γ_k is the clearance rate (enzymatic degradation, typical half-life) - $U_k(\mathbf{v})$ is the local uptake rate (proportional to glial density at that voxel) - $S_k(\mathbf{v}, t)$ is the production term (sum over neurons producing chemical k at this location)

We discretize with forward Euler at a 100 ms chemistry tick:

$$c_k(\mathbf{v}, t + \Delta t_c) = c_k(\mathbf{v}, t) + \Delta t_c \left[D_k(\nabla^2 c_k)(\mathbf{v}, t) - (\gamma_k + U_k(\mathbf{v}))c_k(\mathbf{v}, t) + S_k(\mathbf{v}, t)\right]$$

The 3D Laplacian ∇^2 is implemented as a 7-point stencil:

$$(\nabla^2 c)(\mathbf{v}) = \frac{1}{h^2}\left(\sum_{\text{6 neighbors}} c(\mathbf{v}') - 6c(\mathbf{v})\right)$$

This is the simplest and most cache-friendly 3D discretization and maps directly to a GPU compute shader. The 10 Hz update rate is sufficient for the chemistry timescales (most chemicals have time constants of seconds; 100 ms updates resolve dynamics cleanly).

6.5 Production from neurons

Specific neuron populations produce specific chemicals. For example, the dopamine field has sources at neurons tagged as dopaminergic (typically located in a VTA-analog region). When such a neuron fires, it contributes to the dopamine field in the voxel at its position:

$$S_{\text{dopamine}}(\mathbf{v}_n, t) += k_{\text{prod}} \cdot \delta_{\text{spike},n}(t)$$

Different chemicals have different source distributions: - Dopamine: VTA-analog (sparse, a few hundred neurons) - Serotonin: raphe-analog (sparse) - Adrenaline: locus-coeruleus-analog (very sparse, few dozen neurons) - Cortisol: HPA-analog, responds to sustained threat signal at longer timescales - Oxytocin: hypothalamus-analog, responds to bonding signal - Acetylcholine: basal forebrain analog, tied to arousal and attention - GABA-peptide: distributed across inhibitory interneurons - Melatonin: pineal-analog, circadian-driven

The production rate k_{prod} per spike is small; sustained chemical elevation requires sustained firing of the producing population or many producing neurons firing briefly.

6.6 Receptor binding and neural effect

The mechanism by which chemistry affects neurons is specified in Section 4.6: each neuron samples local chemistry via trilinear interpolation, multiplies by its receptor expression vector, and uses the resulting effect vector to modulate gain, threshold, and plasticity rate.

This is chemistry's primary computational effect. The secondary effect — on body state and through it back into chemistry — is described in Section 10.

6.7 PAD as readout

A central design choice: emotion representations are derived from chemistry, not stored as variables. The pleasure-arousal-dominance (PAD) vector is computed as:

$$P = \text{sigmoid}(w_P^{(\text{D})}\bar{c}_D + w_P^{(\text{S})}\bar{c}_S - w_P^{(\text{C})}\bar{c}_C + \ldots)$$

where $\bar{c}_X$ is the mean concentration of chemical X across the substrate volume, and $w_P^{(X)}$ are tuning coefficients. Similar formulas apply for arousal and dominance. The specific weights are calibrated such that the PAD readout under known chemistry configurations matches the expected values (high dopamine and serotonin, low cortisol → high pleasure; high adrenaline → high arousal; etc.).

Discrete emotion labels (joy, fear, anger, etc.) are assigned post-hoc as the nearest prototype in a space that includes PAD coordinates plus a summary of the chemistry vector. The emotion label is never an input to the substrate; it is an output layer for downstream consumption.

6.8 Computational cost

Chemistry diffusion at 10 Hz over 2.1 M voxels with 8 chemicals: per update, $2 \times 10^6 \times 8 \times (\sim 15 \text{ FLOPs}) = 2.4 \times 10^8$ FLOPs, or at 10 Hz, 2.4×10^9 FLOPs/sec (2.4 GFLOPs/sec). Well within GPU capability.

Production and uptake add $O(N_{\text{producer neurons}} + N_{\text{voxels}})$ per update, roughly an order of magnitude less than diffusion itself.

Receptor coupling at the neuron level (Section 4.6) adds $O(N_{\text{neurons}} \times N_{\text{chem}})$ per frame, ~$250\text{k} \times 8 \times 60 = 1.2 \times 10^8$ FLOPs/sec (0.1 GFLOPs/sec), which is negligible.

Chemistry is therefore cheap relative to neuron integration and synapse propagation, despite being a full 3D field simulation. This is a direct consequence of the unified memory architecture and the cache-friendliness of stencil operations on modern GPU hardware.

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7. Plasticity and Learning

7.1 Design principles

Plasticity in the substrate is:

- **Continuous**: runs every tick. Not gated to a training mode.
- **Chemistry-modulated**: the third factor in our three-factor rule is local chemistry, not a global modulator variable.

- **Compartment-specific**: plasticity rules differ between apical and basal dendrites and between excitatory and inhibitory synapses.
- **Bounded**: metaplasticity (BCM) and glial scaling prevent runaway.
- **Structural**: synapses can be created and destroyed, not just reweighted.
- **Sleep-modulated**: plasticity is gated differently in wake, NREM, and REM.

This section specifies the synaptic weight rules. Structural plasticity is Section 8; sleep regulation of plasticity is Section 9.

7.2 STDP traces

Each synapse maintains pre- and post-synaptic traces:

$$\frac{dx_{\text{pre}}}{dt} = -\frac{x_{\text{pre}}}{\tau_{\text{pre}}} + \delta_{\text{pre-spike}}(t)$$

$$\frac{dx_{\text{post}}}{dt} = -\frac{x_{\text{post}}}{\tau_{\text{post}}} + \delta_{\text{post-spike}}(t)$$

Typical $\tau_{\text{pre}} = \tau_{\text{post}} = 20$ ms, matching cortical STDP measurements (Bi and Poo, 1998).

When a pre-synaptic spike arrives at the synapse, the post-trace x_{post} is sampled: high post-trace means the post-synaptic neuron recently spiked, so pre-post ordering is temporally correct for LTD. When a post-synaptic spike fires, the pre-trace at each incoming synapse is sampled: high pre-trace means pre-before-post, correct for LTP.

7.3 Eligibility trace

Unlike classical STDP which updates weights directly on spike pairs, our rule accumulates eligibility:

On pre-spike at synapse s:

$$e_s \leftarrow e_s + A_{\text{post}} \cdot (-x_{\text{post}}(s)) \cdot f_{\text{compart}}(s)$$

On post-spike at post-neuron n, for each incoming synapse s:

$$e_s \leftarrow e_s + A_{\text{pre}} \cdot x_{\text{pre}}(s) \cdot f_{\text{compart}}(s)$$

where $A_{\text{pre}}, A_{\text{post}}$ are the LTP and LTD amplitude factors (typical values 0.005 and -0.00525), and $f_{\text{compart}}(s)$ is a compartment-specific modulator (typically 1 for apical/basal excitatory synapses, slightly different values for inhibitory synapses onto soma).

The eligibility also decays:

$$\frac{de_s}{dt} = -\frac{e_s}{\tau_e}$$

with $\tau_e \approx 1$ s. This gives the substrate a short-term "memory" of recent spike pairs that can later be converted to weight changes by chemistry.

7.4 Three-factor weight update

At every chemistry tick (10 Hz), eligibility is converted to weight change, gated by local chemistry:

$$\Delta w_s = \eta \cdot M(\mathbf{c}_{\text{local},s}) \cdot e_s \cdot B(\text{pre-rate}, \text{post-rate})$$

where:

- η is the base learning rate (typical value 0.01)
- $M(\mathbf{c})$ is the chemistry modulator, computed from local chemistry at the synapse:

  $$M(\mathbf{c}) = w_D^{\text{plast}} c_{\text{dopamine}} + w_A^{\text{plast}} c_{\text{acetylcholine}} - w_C^{\text{plast}} c_{\text{cortisol}}$$

  with typical weights $w_D^{\text{plast}} = 1.0, w_A^{\text{plast}} = 0.6, w_C^{\text{plast}} = -0.5.$ High dopamine and acetylcholine promote learning; high cortisol suppresses it.
- $B(\cdot)$ is the BCM metaplasticity factor (see below).

The full weight update is applied as:

$$w_s \leftarrow w_s + \Delta w_s$$

followed by bounds $[w_{\text{min}}, w_{\text{max}}]$ (typically 0 and $2g_{\text{max,base}}$).

7.5 BCM metaplasticity

To prevent runaway potentiation, we apply BCM-style metaplasticity (Bienenstock, Cooper, and Munro, 1982). Each neuron maintains a sliding estimate of its own firing rate $\bar{r}_n$ over a long time constant ($\tau_{\text{BCM}} \approx 60$ s):

$$\tau_{\text{BCM}} \frac{d\bar{r}_n}{dt} = r_n - \bar{r}_n$$

The BCM factor for plasticity at synapses into neuron n is:

$$B_n = \phi(r_n - \theta_n)$$

where $\theta_n \propto \bar{r}_n^2$ is the sliding threshold and ϕ is a nonlinearity that is negative below threshold (producing LTD) and positive above (producing LTP). The key property: a neuron that has been chronically active has high θ_n, making further potentiation harder and depression easier. A neuron that has been chronically quiet has low θ_n, making potentiation easy.

This is a homeostatic mechanism that protects against cascade-like all-neurons-saturate failures, which are otherwise a characteristic failure mode of uncontrolled STDP in large networks.

7.6 Compartment-specific plasticity

Plasticity rules differ across compartments:

Apical dendrite synapses: subject to the standard three-factor rule above, but gated additionally by the NMDA plateau state. Synapses contributing to plateau events experience enhanced plasticity, capturing the experimental finding that dendritic plateaus are required for certain forms of LTP (Golding, Staff, and Spruston, 2002).

Basal dendrite synapses: standard three-factor rule.

Perisomatic inhibitory synapses: plasticity is slower and governed by different dynamics. We use a simple inhibitory homeostatic rule (Vogels et al., 2011): inhibitory weights adjust to keep post-synaptic firing at a target rate. This is essential for maintaining E/I balance.

Gap junctions: no plasticity in the current design. (Gap junction plasticity exists in biology but is slow and not well-characterized computationally.)

7.7 The third-factor chemistry in detail

The chemistry modulator $M(\mathbf{c})$ in the three-factor rule encodes the substrate's credit assignment signal. Worth examining concretely:

- A reward-adjacent event produces a dopamine burst at the relevant producer neurons. Dopamine diffuses through the substrate volume. Synapses active during the window when local dopamine is elevated convert their eligibility to weight change; synapses far from the dopamine cloud do not.
- An attention-requiring event elevates acetylcholine in specific regions. Same mechanism: plasticity is enhanced where ACh is high.
- A high-cortisol stressful event suppresses plasticity in regions where cortisol diffuses. This captures the well-documented effect of acute stress on hippocampal LTP (Kim and Diamond, 2002).

The spatial pattern of plasticity at any moment is therefore the intersection of eligibility (which synapses recently had spike pairs?) and local chemistry (which of those synapses are in regions currently being reinforced?). This is fundamentally different from global-modulator approaches and produces different credit assignment behavior.

7.8 Short-term plasticity

In addition to the STDP-based long-term plasticity, synapses exhibit short-term dynamics via the vesicle pool and release probability mechanisms (Section 5.3). These produce effective synaptic weight changes at the seconds timescale that do not persist but that substantially affect ongoing dynamics. Short-term plasticity is fully emergent from the synapse model; no additional rules are required.

7.9 Plasticity telemetry

The substrate exposes plasticity-related metrics continuously:

- Global LTP and LTD rates (weight changes per second, separated by sign)
- Per-region plasticity rates
- Eligibility pressure (total outstanding $|e_s|$ across the substrate, indicating "pending learning" waiting for chemistry gating)
- BCM threshold distribution
- Weight distribution statistics (mean, variance, tails)
- Per-chemistry-modulator contribution (how much learning is gated by dopamine vs acetylcholine vs cortisol?)

These metrics make the learning process legible. An operator should be able to look at a dashboard and see learning happening in real time, see which regions are most plastic, and see what chemistry is gating the plasticity.

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8. Structural Plasticity and Unbounded Growth

8.1 Why structural plasticity matters

A fixed-topology network is a network whose computational capacity is capped at design time. Biological networks, particularly over developmental and learning timescales, are structurally plastic: they add and remove synapses continuously, add new neurons in specific regions throughout life, and reorganize connectivity in response to experience.

For a long-running embodied agent, this matters both theoretically and practically. Theoretically: fixed topology implies fixed representational capacity, which is at odds with the goal of an agent that can acquire new categories of experience over months of operation. Practically: structural plasticity provides a mechanism for the substrate to self-tune its resource allocation, growing capacity where it is needed and freeing resources where it is not.

We implement four forms of structural plasticity: synaptogenesis, synaptic pruning, neurogenesis, and apoptosis. All four run continuously; large structural changes occur during sleep phase transitions when the substrate is in a well-defined quiescent state. The rates are limited per sleep cycle to prevent destabilizing changes, but there is no architectural cap on the total number of neurons or synapses.

8.2 Synaptogenesis

New synapses form between neurons that are not currently connected but whose joint activity suggests they should be. The growth probability at each sleep cycle is:

$$P(\text{grow}(i,j)) = P_0 \cdot c_{i,j} \cdot a_{i,j} \cdot d_{i,j} \cdot g_{i,j}$$

where: - P_0 is a base rate (small, typically 10^{-6} per sleep cycle) - $c_{i,j}$ captures co-activity: how often did i and j fire in close temporal proximity during the preceding wake period? - $a_{i,j}$ is the chemistry alignment: how similar were the local chemistry states at the two neurons' positions? - $d_{i,j}$ is the distance factor: $e^{-\|\mathbf{x}_i - \mathbf{x}_j\|/\lambda_{\text{grow}}}$ - $g_{i,j}$

is an age factor: recently-born neurons secrete growth factors that bias connectivity toward them

When a synapse is created, it inherits the pre-synaptic neuron's E/I identity (preserving Dale's law), is assigned a synapse type based on cell types, receives an initial small weight, and receives a delay based on distance and conduction velocity.

8.3 Synaptic pruning

Synapses are pruned when their contribution to network function is low. Criteria:

- **Weight-based**: synapses whose absolute weight has been below a small threshold for an extended period (multiple sleep cycles) are pruned.
- **Spine size-based**: the spine size s is a structural proxy that tracks recent activity and weight; when it falls below a threshold, the synapse is pruned.
- **Microglial-analog flagging**: during deep sleep, a microglial-analog process scans synapses and probabilistically flags under-utilized ones for removal.

Pruning rates are bounded per sleep cycle (typically at most a few percent of total synapses per cycle) to prevent mass disconnection events.

8.4 Neurogenesis triggers

New neurons are born during sleep phase transitions, specifically at the NREM-to-REM boundary where structural plasticity is globally active. Four triggers:

Local saturation pressure. A region with many synapses near maximum weight and neurons firing near their metabolic ceiling signals that the region is at capacity. A new neuron is instantiated at a position in that region.

$$\text{SaturationPressure}(\text{region}) = \frac{\#\{s : w_s > 0.9 w_{\max}\}}{\#\{s\}} + \frac{\#\{n : \alpha_n < 0.3\}}{\#\{n\}}$$

When this exceeds a threshold, neurogenesis is triggered.

Novel pattern pressure. If an input pattern arrives that existing neurons do not discriminate (high reconstruction error from hippocampal-analog region; high lateral competition without a clear winner), a new neuron is born with connectivity biased toward representing the pattern.

Developmental schedule. During specific substrate ages — early after initialization, and during reproductive-module-driven pubertal profiles (if enabled) — the baseline neurogenesis rate is elevated.

Replacement. When a neuron is pruned and the region's activity statistics suggest capacity is still needed, a replacement is spawned at a nearby position.

Each new neuron is born with: - Position in 3D space (within the target region) - Cell type drawn from the region's characteristic distribution - Parameters within the normal variance for that cell type - Empty initial connectivity (no synapses yet; these are grown over subsequent sleep cycles) - Elevated growth factor secretion (biases nearby synaptogenesis toward itself for a transient period) - Baseline ATP level (must be supported by local glial supply) - Receptor expression vector drawn from the region's typical profile - Fresh generational ID

8.5 Apoptosis triggers

Neurons are removed when they fail to contribute. Five triggers:

Metabolic insufficiency. If ATP has been chronically below a threshold across multiple sleep cycles despite normal sleep-dependent replenishment, the neuron is marked for removal. This captures silent neurons (not firing enough to justify their metabolic cost) and pathologically hyperactive neurons (exceeding their metabolic budget).

Consolidation redundancy. During sleep consolidation, if a neuron's representational contribution is well-covered by others (measured as the fraction of its spike patterns that are reconstructable from other neurons), it is probabilistically pruned.

Critical-period closure. Neurons born in a given sleep cycle have a critical period during which they must integrate into a stable assembly. Measured by clustering coefficient in their local connectivity graph. If this is below threshold at the end of the critical period, the neuron is pruned.

Microglial elimination. Under-connected neurons (low summed $|w|$ across outgoing synapses) are probabilistically flagged during deep sleep.

Resource pressure. When the substrate operates near its metabolic ceiling, resource competition selects low-contribution neurons for elimination. This is the "use it or lose it" mechanism at the population level and is the mechanism through which the substrate self-regulates to its hardware envelope.

When a neuron dies: - All outgoing synapses are removed, their state slots returned to the free pool - All incoming synapses (now targeting a dead neuron) are similarly pruned - The neuron's state slot is returned to the free pool - Its ID is retired: the slot's generation counter is incremented, so any held references to the old ID become stale

8.6 Generational indexing

The substrate uses **generational IDs** to allow slot reuse without aliasing:

```
NeuronId = (slot_index: u32, generation: u32)
```

When a neuron dies, its slot's generation is incremented. A new neuron can then be placed in the same slot with the new generation. Any external reference still holding the old ID will see a generation mismatch and correctly report that the referent is dead.

The same mechanism applies to synapses. This solves the identity problem cleanly without requiring garbage collection or reference counting.

8.7 Metabolic competition

Growth rates and apoptosis rates are coupled through a shared resource: glial metabolic support. Each glial cell has a finite supply capacity, distributed to nearby neurons. When more neurons exist in a region than the glia can support, per-neuron ATP supply falls, increasing apoptosis probability for low-contribution neurons and slowing neurogenesis.

Mathematically:

$$\text{per-neuron ATP supply at } \mathbf{x} = \frac{\sum_g G(g) \cdot K(\mathbf{x}, \mathbf{x}_g)}{\sum_n K(\mathbf{x}, \mathbf{x}_n)}$$

where the numerator sums glial capacity weighted by kernel K (spatial radius of support) and the denominator sums neuron count in the same way. More neurons in a region means less ATP per neuron, which provokes apoptosis of the weakest until the density matches the supply.

This produces the right qualitative behavior: the substrate approaches its hardware envelope gracefully through a soft carrying capacity, not through a hard cap. The carrying capacity is set by the glial density and the hardware resources; if both scale up, so does the population ceiling. If neither does, the population stabilizes.

8.8 Rate limits

Per sleep cycle, the rates of structural change are bounded:

Operation	Per-cycle rate limit
Neurogenesis	0.1% – 1% of current population
Apoptosis	Same
Synaptogenesis	1% – 5% of current synapse count
Synaptic pruning	Same

These limits prevent mass restructuring events that could destabilize the substrate. The default rates are at the low end of these ranges; higher rates are appropriate for developmental bootstrapping but not for mature substrates.

8.9 Identity and structural change

A substrate that constantly adds and removes neurons raises a question: what does it mean for the agent's identity to persist?

The answer is that identity is a property of the network as a whole, not of specific neurons. What is preserved across normal structural turnover:

- Large-scale connectivity statistics
- Long-term weight patterns that have consolidated into distributed representations

- Chemistry receptor profiles across regions
- Durable cell assemblies (which can survive member turnover if consolidated properly)

What is not preserved: the identity of specific neurons, the specific synapses. This matches the biological situation: real brains replace most of their molecular contents over years while the person remains the person.

Practical consequence: substrate snapshots (Section 13) do not attempt to preserve specific neuron identities across major time gaps. They preserve state at a point in time, and loading a snapshot restores that state.

---

9. Sleep and Consolidation

9.1 Sleep as a regime change

In the substrate, sleep is not a state the agent enters — it is a chemistry regime change whose consequences include the behaviors we associate with sleep (reduced external responsiveness, memory consolidation, restoration). The chemistry regime is the primary state; the behavioral consequences follow.

Chemistry regimes:

Phase	Acetylcholine	Adenosine	Adrenergic tone	Network behavior
Wake	High	Low (rising)	Moderate	High plasticity, feedforward-dominant
Light sleep	Declining	Rising	Declining	Plasticity reducing, spindle-like oscillations
NREM (deep)	Low	High	Low	Slow oscillations, sharp-wave ripples, consolidation
REM	High	High	Very low	Reverberant, internal input, active plasticity
Waking	Rising	Clearing	Rising	Return to wake dynamics

The state machine transitions are driven by adenosine accumulation, circadian signaling (via melatonin), and external conditions. The state machine itself is simple; the interesting dynamics are consequences of the chemistry regime.

9.2 Memory consolidation mechanisms

During NREM: - Hippocampal-analog region produces sharp-wave ripple-like events (100–200 Hz bursts) - Ripples replay recent activity patterns with time compression (faster than wake) - Low acetylcholine during NREM gates plasticity selectively: only strongly-activated synapses undergo LTP - Neocortical-analog regions receive the replayed activity and gradually integrate it

This is the standard hippocampal-to-neocortical consolidation pathway (McClelland, McNaughton, and O'Reilly, 1995; Kumaran, Hassabis, and McClelland, 2016). Its implementation in the substrate requires:

- A distinct hippocampal-analog region with fast plasticity (high learning rate during wake; selective consolidation during sleep)
- A distinct neocortical-analog region with slow plasticity (high capacity, slow to change, integrates over many exposures)
- A mechanism for ripples: recurrent excitation in the hippocampal-analog region, gated by cholinergic state

All of these are natural consequences of the substrate's design: zoned populations with differential plasticity rates, recurrent connectivity in specific regions, chemistry-gated plasticity. We do not need special consolidation code — we need the chemistry regime to produce ripples during NREM, and consolidation happens.

9.3 Synaptic homeostasis during sleep

We implement a form of the SHY (Synaptic Homeostasis) hypothesis (Tononi and Cirelli, 2014): during deep sleep, glial cells perform region-wide synaptic downscaling:

$$w_s \leftarrow w_s \cdot (1 - \delta_{\text{scale}})$$

with δ_{scale} small (typically 0.02 per deep sleep episode). All synapses in the region are scaled. This prevents the substrate from saturating during days of wake plasticity.

Crucially, this downscaling preserves relative strengths: consolidated patterns remain stronger than noise patterns after downscaling, even though absolute values drop. Combined with the selective replay-driven LTP during ripples, the net effect is that important patterns get reinforced while background noise gets scaled down.

9.4 REM and dream content

During REM, the substrate runs in a reverberant mode with external input suppressed. The chemistry profile (high ACh, low noradrenaline and serotonin) corresponds to wake-like plasticity with dream-like representational content.

The spontaneous activity during REM is biased by recent eligibility: synapses with high unresolved eligibility from the preceding wake period are more likely to activate during replay. This produces dream content that reflects the day's experience, weighted by emotional salience (since high chemistry modulation during wake created more eligibility).

If a large language model is available in the cognitive layer, it can be invoked during REM to **narrate** the substrate's internal activity into natural-language dream content, which can be logged to a journal. The substrate's activity is the seed; the LLM is the narrator. The LLM does not drive the substrate's activity during REM.

9.5 Sleep drive

Sleep onset is driven by adenosine accumulation and circadian state. Adenosine is a byproduct of ATP metabolism:

$$\frac{dc_{\text{adenosine}}}{dt} = k_{\text{ATP-decay-rate}} \cdot \big(\text{average spiking activity}\big) - \gamma_{\text{adenosine}} c_{\text{adenosine}}$$

During wake, adenosine accumulates. When it crosses a threshold and the circadian melatonin phase permits, the substrate transitions to light sleep. During sleep, adenosine clears (both via reduced production and via increased clearance rates).

Sleep deprivation is therefore a real substrate pressure: accumulated adenosine degrades neuron performance (via the ATP supply mechanism), plasticity is impaired at high cortisol (which tends to be high under sustained wake), and mood baseline drifts in observable ways.

---

10. Interoception and Embodiment

10.1 Why embodiment matters

Following interoceptive inference theory (Seth, 2013; Barrett, 2017), emotion in biological organisms is not a label assigned by the brain — it is the brain's readout of body state combined with predictive inference about the causes of that state. An embodied agent that reports "being afraid" is reporting on a brain-body configuration, not stating a belief.

For a synthetic agent, this implies that truly lifelike affective behavior requires a body-brain loop. Without one, emotion is a story told about activity; with one, emotion is a readout of a real closed loop. The substrate's chemistry field is one half of this loop; the interoception module is the other half.

The agent is not physically embodied in hardware (there is no robot); the body is simulated. What matters is not the physical reality but the loop: chemistry affects body, body feeds back into chemistry, and the agent's internal state depends on this circular dynamic rather than on externally set variables.

10.2 Body state

The simulated body maintains the following state at 10 Hz:

- Heart rate (bpm) and heart rate variability (HRV, as RMSSD-analog)
- Respiration rate and respiration cycle phase
- Gut activity (a slow oscillator with stress-responsive amplitude)
- Core temperature (small variations)
- Muscle tension
- Interoceptive noise baseline

Each variable has its own dynamics and coupling. For example, heart rate:

$$\tau_{\text{HR}} \frac{d\text{HR}}{dt} = \text{HR}_{\text{target}}(\mathbf{c}) - \text{HR} + \xi_{\text{HR}}(t)$$

where $\text{HR}_{\text{target}}(\mathbf{c})$ depends on chemistry:

$$\text{HR}_{\text{target}} = 70 + 30 \cdot \tanh(w_A^{\text{HR}} \bar{c}_{\text{adrenaline}} + w_D^{\text{HR}} \bar{c}_{\text{dopamine}} - w_M^{\text{HR}} \bar{c}_{\text{melatonin}})$$

High adrenaline or dopamine pushes heart rate up; high melatonin pushes it down. Similar formulas for the other body variables.

10.3 Body-to-chemistry feedback

The body state feeds back into chemistry via dedicated production pathways:

- Sustained high heart rate → cortisol production elevated (body signals "we're stressed")
- Low HRV → sympathetic bias → noradrenaline production elevated
- Gut activity oscillations → vagal-analog signal → serotonin modulation
- Slow respiration sustained → parasympathetic signal → GABA-peptide elevation
- Circadian phase → melatonin production modulated

This is the closing side of the loop. The body is driven by chemistry; the body drives chemistry. Under steady external conditions, the loop settles into a resting state. Under perturbation (a threatening event, a rewarding event, a sustained interaction), the loop transitions to a new configuration that persists on the timescales of the slowest components.

10.4 Interoceptive neurons

A specific population of neurons in the substrate receives direct projections carrying body state signals. These **interoceptive neurons** have their I_{intero} term (Section 4) driven by body variables:

$$I_{\text{intero},n} = \sum_k w_{n,k} \cdot b_k(t)$$

where $b_k(t)$ is body variable k and $w_{n,k}$ is the coupling weight. Interoceptive neurons form an "insula-analog" population whose activity tracks body state and projects into chemistry production regions, completing the loop within the substrate itself.

10.5 Observable consequences

The loop produces behaviors that a stateless inference system cannot replicate:

- Mood persists. A stressful event in the morning leaves the body in an elevated state (high HR, low HRV, elevated gut activity) that maintains cortisol production for hours. PAD readout reflects this persistence.
- Circadian rhythm shows through. Heart rate, respiration, muscle tension all vary with time of day because circadian chemistry drives them.
- Recovery is slow. After a threatening event, full return to baseline takes time because the body must return to baseline and the chemistry must clear.
- Acute responses are coherent. An acute adrenergic event produces a tight cluster of coordinated body-and-chemistry changes that match biological autonomic responses.

These observable consequences are, we argue, what makes the system feel lifelike. Without the loop, emotion is a declared label; with it, emotion is a joint state of substrate, body, and chemistry that takes time to enter and time to leave.

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11. Developmental Bootstrapping

11.1 Why development matters

A substrate that depends on activity-dependent self-organization cannot be shipped with fully-formed topology. The topology at maturity is a consequence of early activity patterns, growth factor distributions, and critical-period plasticity schedules — things that cannot be hand-designed without losing the emergence properties that make the substrate interesting.

At the same time, a production agent cannot wait for the substrate to mature in real time. Biological brains take months to years to develop the functional structures we want to evoke. For deployment, we need a mechanism to fast-forward development: produce a mature substrate from a small seed specification, in a time budget of hours rather than years.

This section describes the developmental bootstrapping process and the critical-period scheduling that governs it.

11.2 Seed specification

A substrate is specified by a seed configuration that includes:

- Volume dimensions (physical size of the substrate space)
- Voxel grid resolution
- Initial neuron count per region
- Per-region cell type distribution
- Initial chemistry baseline
- Chemistry source neuron specifications (which cells produce which chemicals)
- Seed connectivity rules (distance biases, cell-type-pair probabilities)

- Growth factor distributions (to bias early developmental connectivity)
- Critical-period schedule (Section 11.4)
- Glial distribution
- Interoceptive neuron assignments
- RNG seed for reproducibility

The seed specification is compact — typically a few kilobytes of YAML. It does not include specific neuron identities, specific synapses, or specific weights. It describes the initial conditions and the rules, and lets the developmental process produce the substrate.

11.3 Developmental phases

Development proceeds through phases with distinct plasticity regimes:

Phase D0: Placement. Neurons are placed in the substrate volume according to the seed specification. Cell types assigned. Positions set. No connectivity yet. Glial cells placed. Initial chemistry field populated with baseline concentrations and any seeded gradients.

Phase D1: Seed connectivity. Initial synapses formed according to distance-biased, cell-type-pair-biased probability rules. This produces a sparse initial topology. No learning happens; synapses are given small initial weights. Growth factor fields are active, biasing subsequent connectivity.

Phase D2: Spontaneous activity. The substrate generates spontaneous activity: endogenously driven, not externally clamped. This is analogous to spontaneous retinal waves and cortical activity waves observed in embryonic development (Blankenship and Feller, 2010). Specific populations produce burst-like activity that propagates through the seed topology. Plasticity is on, with elevated learning rate and aggressive structural plasticity (synaptogenesis rate 10–50x the mature rate).

During D2, the topology rapidly reorganizes. Correlated activity patterns strengthen specific connections; uncorrelated activity fails to strengthen. Populations that fire together begin to wire together even if initial connectivity was sparse. Cell-assembly formation starts.

Duration of D2 in accelerated mode: approximately 4–8 hours of simulated time at 10x real-time (i.e., 24–48 minutes wall clock). In real-time mode: days to weeks.

Phase D3: Input-driven refinement. External input is introduced (either from a scripted bootstrapping input stream, or from live sensors if the agent is already running). The substrate receives structured input for the first time. Plasticity is still elevated but gradually decreases. Functional specialization emerges: regions that receive consistent input types begin to specialize.

This is the phase where region-specific behaviors emerge. The region that receives interoceptive input develops into something insula-like. The region that receives hippocampal-style rapid-encoding input develops hippocampal-like behavior. Not because these were programmed, but because the chemistry regime, connectivity rules, and input statistics combine to produce functional specialization.

Duration: several to many hours at 10x real-time; several days in real-time.

Phase D4: Consolidation and stabilization. Learning rate gradually decays to adult values. Structural plasticity rates drop. The substrate enters approximately its mature regime. Some ongoing plasticity continues (mature substrates are never fully static), but the character of change transitions from rapid reorganization to slow refinement.

Phase D5: Deployment. The substrate is considered mature and is snapshotted as an "adult substrate" for production use. Further development continues but the substrate is ready to deploy.

11.4 Critical-period scheduling

Key plasticity parameters are scheduled as functions of developmental age:

Parameter	D0	D1	D2	D3	D4	D5
Base learning rate multiplier	0	0.1	5.0	2.0	1.0	1.0
Synaptogenesis rate multiplier	0	1.0	20.0	5.0	2.0	1.0
Pruning rate multiplier	0	0.1	5.0	3.0	1.5	1.0
Neurogenesis rate multiplier	0	0	5.0	2.0	1.0	1.0
Critical period openness	—	full	full	closing	mostly closed	local only

Critical periods in biology are windows during which specific circuits can be shaped by input but become progressively less plastic over time. In the substrate, we implement per-region closing schedules: the chemistry regulator controls critical-period openness via acetylcholine baseline levels, which in turn modulate plasticity. During D2–D3, ACh baseline is elevated globally; during D4 it drops; during D5 it varies locally based on need.

11.5 Reproducibility and seeding

A given seed configuration with a given RNG seed produces a deterministic developmental trajectory (assuming deterministic hardware behavior, which requires care on GPUs). This enables:

- Reproducible research substrates (same seed → same mature substrate)

- A/B testing of seed variations
- Debugging: replay a developmental trajectory to isolate when specific behaviors emerged

Production substrates use a canonical seed for a reference “adult substrate” that can be distributed as a starting point. Users who want different initial character can use a different seed.

11.6 The adult substrate

The output of D5 is an adult substrate: a mature, serialized substrate state that can be deployed as a production agent. This is what users would typically install and run rather than starting from a seed themselves.

Adult substrates are: - Fully serializable (using the snapshot format, Section 13) - Portable across hardware (with appropriate backend support) - Point-in-time captures that still undergo continued learning during deployment

A canonical adult substrate is the foundation for production deployments. Users receive a known-good starting point and the substrate develops further through interaction.

11.7 Pre-natal substrates

For research purposes, substrates can be observed at any developmental phase. A “pre-natal” substrate from phase D2 is a research object in its own right — it shows the substrate’s self-organization from minimal initial conditions. Such substrates may not exhibit stable function but are interesting for studying how function emerges.

11.8 The time-scale compression question

Accelerating development from biological timescales to minutes raises a question: is the fast-forwarded substrate qualitatively equivalent to a substrate that developed over the full biological timescale? We believe approximately yes but cannot guarantee exact equivalence. The relevant factors:

- Chemistry field dynamics scale with timestep size; at 10x acceleration, diffusion coefficients are scaled so spatial profiles remain consistent
- Plasticity rules scale analogously
- Stochastic variability is preserved (RNG calls scale with simulated time)

What is not preserved at high acceleration: any behavior that depends on slow molecular dynamics not captured in our model. Since we don’t model molecular dynamics explicitly (our chemistry is coarse-grained concentration fields), this is not an issue in practice.

We recommend: production adult substrates are generated at 1–10x acceleration to balance realism and deployment speed. Research substrates targeting specific phenomena should be generated at whatever rate best captures the phenomena under study.

12. Operator Telemetry and Substrate Legibility

12.1 Why legibility matters

A substrate operating as the backbone of a long-running agent is opaque by default. Without dedicated instrumentation, operators cannot tell whether the substrate is healthy, learning productively, or drifting toward pathology. Debugging is worse: a "wrong" behavior could originate anywhere — chemistry, plasticity, structural changes, body state, or upstream cognitive logic.

We treat legibility as a first-class design requirement. Every mechanism in the substrate emits telemetry; every telemetry stream has a dashboard view; every unusual event raises an alert. An operator should be able to answer, at a glance and at any time:

- Is the substrate healthy?
- What is it learning?
- What chemistry is driving its behavior?
- What is its body state telling us?
- Are structural changes happening at expected rates?
- Is any mechanism drifting toward failure?

This section specifies the telemetry architecture.

12.2 Metric categories

Telemetry is organized into eight categories:

1. Substrate health. Firing rate distributions, E/I balance, criticality σ, ATP distribution, synaptic weight distribution tails. Intended to answer "is the substrate in biologically reasonable regimes?"

2. Plasticity. Global and per-region LTP/LTD rates, eligibility pressure, BCM threshold distributions, chemistry modulator contributions, compartmental plasticity rates. Intended to answer "what is being learned where?"

3. Chemistry. Per-chemical global mean and variance, spatial gradient strength, production and uptake rates, receptor saturation indicators. Intended to answer "what modulators are active and where?"

4. Structural. Current neuron and synapse counts, neurogenesis and apoptosis rates, synaptogenesis and pruning rates, per-region population dynamics, spine size distributions. Intended to answer "how is the substrate's physical structure changing?"

5. Interoception. Body variable trajectories, body-chemistry coupling metrics, autonomic balance indicators. Intended to answer "what is the simulated body telling the substrate?"

6. Sleep. Current sleep phase, phase history, adenosine accumulation, circadian phase, replay event frequency, consolidation throughput. Intended to answer "how is the sleep cycle progressing?"

7. Performance. Per-phase tick latency (p50, p95, p99), frame drop rate, GPU utilization, memory footprint, thread pool queue depths. Intended to answer "is the substrate meeting its 60 Hz contract?"

8. Meta. Snapshot rate, journal write rate, substrate age, events processed, telemetry overhead. Intended to answer "how is the substrate infrastructure itself performing?"

Each category has its own dashboard panel, its own alert thresholds, and its own expected-range documentation.

12.3 Visualization requirements

Some substrate dynamics are difficult to understand from time-series metrics alone. The architecture includes:

3D chemistry field visualizer. Interactive slice and volume renderer for the chemistry voxel grid. Operators can see where dopamine is elevated, how cortisol spatial gradients evolve over time, which regions have active neuromodulators. Essential for understanding credit assignment events.

Connectivity graph viewer. Abstract representation of substrate connectivity, updated from structural changes. Shows which regions are strongly connected, which are sparsely connected, how connectivity changes over time.

Substrate timeline. Horizontal time-axis view showing sleep phases, structural events (birth, death bursts), chemistry extremes, and significant external events on a common timeline. Enables retrospective analysis: "what happened Monday afternoon?"

Neuron inspector. Per-neuron view showing membrane dynamics, spike raster, local chemistry, ATP, plasticity traces. Used for debugging specific cells or understanding what a named cell is doing.

These visualizations are not required for the substrate to function, but they are required for the substrate to be understandable.

12.4 Alert design

Alerts trigger at three severity levels:

Info. Expected but notable events: neurogenesis burst completed, sleep phase transition, named milestone snapshot reached. Logged but does not require operator action.

Warning. Values outside expected range: criticality drifting, chemistry saturation, population growth exceeding rate limit, weight norm outside watchdog band. Requires investigation but substrate continues operation.

Critical. Values in pathological range: sustained criticality violation, weight norm outside emergency band, structural change exceeding safety limit, 60 Hz contract violated for extended period. Triggers protective measures (adaptive quality, reduced plasticity) and alerts operator.

Alert thresholds are configurable and documented. Initial thresholds are derived from the expected-value ranges in Section 12.2.

12.5 Cross-cutting queries

Operators commonly want to answer questions that span multiple telemetry streams:

- "What happened to the substrate when viewer X arrived?" (chemistry × structural × plasticity around a specific time)
- "Why did the substrate's mood shift this morning?" (chemistry × interoception × sleep state)
- "Which of last week's learning has consolidated?" (plasticity trajectories pre- and post- sleep cycles)
- "Are any regions trending toward saturation?" (structural + plasticity per-region)

The telemetry architecture supports these cross-cutting queries via a unified time-indexed metric store. Prometheus handles the high-frequency metrics; a separate columnar time-series store handles the analytical queries.

12.6 Legibility budget

Telemetry has overhead. A substrate that emits metrics every tick for every neuron would drown in its own telemetry. The architecture budgets telemetry tightly:

- Per-tick metrics: aggregated (e.g., rate histograms, not per-event)
- Per-region metrics: every 10 ticks (6 Hz)
- Per-neuron details: only when subscribed (operator has opened a neuron inspector)
- Large visualizations: polled (not pushed)

Total telemetry overhead target: under 0.5 ms per 60 Hz frame, including aggregation and export. Achievable with careful metric design and in-process aggregation.

12.7 Audit logs

Beyond metrics, the substrate logs all structural changes, snapshots, configuration changes, and significant events to an audit log. This is separate from telemetry (which is aggregated and short-retention) and is intended for long-term record-keeping: months or years of substrate history at individual-event granularity.

Audit logs are append-only and integrity-protected (hash chains). They are the definitive record of what happened to a specific substrate instance over its lifetime.

12.8 Operator interface

A dedicated operator dashboard integrates all of the above:

- At-a-glance health summary (one screen)
- Drill-down panels for each telemetry category
- Time-range selection for retrospective analysis
- Interactive chemistry field visualizer
- Neuron inspector
- Alert history
- Audit log viewer
- Snapshot browser

The interface is a web application running locally on the deployment host, accessible over the loopback network interface. It does not require network connectivity beyond local access.

---

13. Implementation

11.1 Language and runtime choice

The substrate is implemented in Python. This is a deliberate choice over lower-level languages (C++, Rust) based on three factors:

Ecosystem alignment. The computational neuroscience and machine learning communities operate in Python. Reference implementations for nearly every mechanism in the substrate exist in Python (Brian2, NEST's Python frontend, snnTorch, PyTorch ecosystem). Staying in Python minimizes friction for lifting ideas from the literature and for comparing against existing systems.

Iteration speed. The substrate has many parameters and many mechanisms. Research and tuning iterations benefit enormously from Python's fast edit-run cycle. A compiled Rust substrate would be ~10x more productive for mature deployed code and ~3x less productive for the exploration and tuning phase.

Acceleration headroom. Python is slow for per-element operations but fast enough for orchestration. The hot paths can be accelerated with Numba JIT, CuPy for GPU array operations, PyTorch for neural network-shaped operations, or direct GPU compute via CuPy/PyTorch custom kernels. Performance-critical paths escape Python; the orchestration stays Python.

The cost of this choice is a 10–20% performance overhead versus a pure-compiled implementation. On target hardware this translates to ~15% fewer neurons at 60 Hz, which we consider acceptable.

11.2 Package structure

The substrate is structured as a Python package with the following modules:

```
substrate/
```

```
├── core/                       # shared types, units, protocols
│   ├── arena.py               # arena-based slotmap for dynamic allocation
│  ├── chunked_soa.py       # chunked structure-of-arrays for neurons/synapses
│   ├── rng.py                  # deterministic RNG per subsystem
│   └── types.py                # NeuronId, SynapseId, VoxelCoord, units
├── neurons/                    # EMCN implementation
│   ├── izhikevich.py           # soma dynamics (Numba JIT + CuPy)
│   ├── compartments.py         # apical/basal/NMDA plateau
│   ├── stochastic_channels.py  # Markov chain gating
│   ├── metabolism.py           # ATP dynamics
│   └── cell_types.py           # taxonomy and parameters
├── synapses/                   # conductance synapses
│   ├── kinetics.py             # dual-exponential + Mg block
│   ├── delay_queue.py          # ring buffers per delay class
│   ├── release.py              # stochastic release, vesicle pool
│   ├── gap_junctions.py        # electrical synapses
│   └── topology.py             # sparse connectivity, Dale's law
├── chemistry/                  # 3D chemistry field
│   ├── field.py                # voxel tensor (CuPy or NumPy)
│   ├── diffusion.py            # 7-point stencil kernel
│   ├── receptors.py            # receptor coupling
│   ├── production.py           # neuron-to-field production
│   └── readout.py              # PAD computation
├── glia/                       # glial cells
│   ├── state.py                # sparse glial population
│   ├── metabolic_support.py    # ATP supply distribution
│   └── scaling.py              # synaptic downscaling
├── plasticity/                 # learning rules
│   ├── stdp_traces.py          # pre/post trace updates
│   ├── eligibility.py          # eligibility trace management
│   ├── three_factor.py         # chemistry-gated weight updates
│   ├── bcm.py                  # metaplasticity
│   └── structural.py           # growth and pruning (sleep-gated)
├── interoception/              # body state and loop
│   ├── body.py                 # body variable dynamics
│   ├── coupling.py             # body-to-chemistry and chemistry-to-body
│   └── receptors.py            # interoceptive neuron projections
├── sleep/                      # sleep state machine
│   ├── state_machine.py        # wake/NREM/REM transitions
│   ├── chemistry_regimes.py    # per-phase chemistry modulation
│   ├── replay.py               # ripple-like events
│   └── dreams.py               # REM orchestration
├── development/                # initial wiring and growth
│   ├── placement.py            # initial neuron placement
│   ├── seed_connectivity.py    # initial connectivity rules
│   ├── growth_factors.py       # developmental chemistry biases
│   └── critical_periods.py     # phase-dependent plasticity
├── persistence/                # snapshots
```

```
│   ├── snapshot.py           # full state serialization
│   ├── journal.py            # incremental WAL
│   ├── versioning.py         # schema migration
│   └── compression.py        # zstd compression pipeline
├── telemetry/                # observability
│   ├── metrics.py            # counters, histograms
│   ├── prometheus.py         # exporter
│   └── tracing.py            # structured traces
├── orchestrator/             # top-level loop
│   ├── step.py               # one-tick driver
│   ├── scheduler.py          # multi-timescale scheduling
│   └── quality.py            # adaptive quality controller
├── accel/                    # acceleration backends
│   ├── numba_backend.py      # JIT-accelerated CPU paths
│   ├── cupy_backend.py       # GPU array operations
│   ├── torch_backend.py      # PyTorch-ROCm backend
│   └── backend_dispatch.py   # runtime backend selection
└── api/                      # external interface
    ├── state_queries.py      # PAD, emotion, metrics readout
    └── events.py             # event injection from cognitive layer
```

Approximately 50 modules, with an estimated ~30,000 lines of implementation code.

11.3 Four-tier acceleration strategy

Hot paths use tiered acceleration. The strategy:

Tier 1 — NumPy. Reference implementation of every mechanism. Correctness-checked. Used for testing, single-neuron simulations, and small-scale validation. Always available regardless of hardware.

Tier 2 — Numba JIT. For CPU hot paths where vectorized NumPy is awkward: delay queue processing, sparse synapse iteration, per-neuron conditionals. Numba provides JIT-compiled ahead-of-time parallel loops with minimal code changes. Used for medium-scale CPU simulation.

Tier 3 — CuPy. For GPU array operations: chemistry diffusion (3D convolution), dense matrix operations in neuron integration, sparse matrix operations in synapse propagation. CuPy is NumPy-compatible and supports AMD ROCm. Used as the primary GPU backend on target hardware.

Tier 4 — PyTorch with ROCm. For operations that benefit from PyTorch's infrastructure: autograd if needed for auxiliary meta-learning, cuDNN-equivalent convolutions, and any path where PyTorch's kernel selection outperforms CuPy. The chemistry diffusion path in particular benefits from PyTorch's optimized 3D convolutions. Also serves as a fallback when CuPy-ROCm has gaps.

The backend is selected at runtime based on available hardware. On target hardware (AMD Strix Halo), the default stack is NumPy for reference + CuPy-ROCm for GPU + Numba for CPU hot paths.

11.4 Dynamic allocation for unbounded substrate

Because the substrate must support unbounded neurogenesis and apoptosis, the storage cannot be fixed-size arrays. Instead:

Arena-based slotmap. Each mutable entity type (neurons, synapses, glial cells) has an arena:

```
class Arena:
    def __init__(self, chunk_size=65536):
        self.chunks: list[StateChunk] = []
        self.free_list: list[int] = []
        self.generation: np.ndarray  # one entry per slot
        self.alive: np.ndarray       # bitmap
        self.count: int = 0
```

Allocation pops from the free list if non-empty; otherwise allocates a new slot (growing chunks as needed). Deallocation marks the slot dead, increments its generation, and adds to the free list.

Chunked structure-of-arrays. Per-neuron state is split across parallel arrays, each chunked for locality:

```
class NeuronState:
    v_soma: ChunkedArray[float32]
    u_soma: ChunkedArray[float32]
    v_apical: ChunkedArray[float32]
    ...
```

Chunked arrays expose a flat view for vectorized operations (neuron integration across all neurons is a single CuPy kernel dispatch). Chunks are allocated and freed dynamically as population grows and shrinks.

Sparse connectivity. Synapses are stored in CSR format keyed by post-synaptic neuron. Addition and removal of synapses reorganizes the CSR structure during sleep cycles when the substrate is quiescent.

Overhead of this dynamic allocation versus fixed arrays: approximately 10–15% per-operation cost for the indirection. Paid only on allocation and deallocation; per-tick substrate compute is unaffected because it operates on the flat chunked views.

11.5 Concurrency model

The substrate runs on an asyncio event loop in a single process. Inside each tick:

1. Event queue drain (synchronous)
2. Neuron integration (synchronous call into GPU; GPU work proceeds asynchronously via CuPy streams)
3. Synapse propagation (overlaps with neuron integration via CuPy streams)
4. Chemistry diffusion (on chemistry-tick frames; GPU)
5. Plasticity updates (GPU)
6. Readout (CPU, trivial)

GPU operations are dispatched from the main thread; their completion is awaited at the tick boundary. CuPy streams allow overlap between neuron and synapse work when they operate on independent data.

Snapshot writes happen on a thread pool, never blocking the main tick. Telemetry exports happen on their own background thread.

Sleep-phase transitions pause normal dynamics briefly for structural changes. This is the only operation that visibly stalls the substrate, and it happens at natural phase boundaries.

11.6 Hardware mapping on target platform

On a consumer workstation with 16-core CPU, integrated GPU, NPU, and unified 128 GB memory:

CPU assignment: - Cores 0–1: 60 Hz main tick loop, event dispatch - Cores 2–3: cognitive layer (planning, LLM orchestration, memory indexing) - Cores 4–7: Numba-accelerated substrate paths (chemistry uptake, glial bookkeeping, some plasticity) - Cores 8–11: background processes (journal writes, snapshot serialization, telemetry) - Cores 12–15: burstable for transient loads (TTS post-processing, LLM prompt preparation)

iGPU assignment: - Substrate hot path (neuron integration, synapse propagation, chemistry diffusion, plasticity): primary workload, owns ~60% of iGPU time on average - Local LLM inference: time-shared, ~20% average, dispatched async from cognitive layer - Text-to-speech: time-shared, ~10% average, dispatched from cognitive layer

NPU assignment: - Speech-to-text (Whisper or equivalent): continuous when audio active - Face analysis, gaze tracking: continuous at 15 Hz when video active - Prosody and feature extraction: continuous at 10 Hz when audio active - Semantic embedding model for memory indexing: background, when idle

Memory layout: - OS + baseline services: ~4 GB - Python process (cognitive, presentation): ~6 GB - Substrate hot state: ~3–8 GB (scales with neuron count) - GPU-resident buffers: ~4–16 GB (unified memory, shared allocation) - LLM weights (quantized to ~45 GB): pinned - TTS models: ~1 GB - Perception models (NPU-accessible): ~2 GB - Vector store for semantic memory: ~4–16 GB - OS page cache and headroom: balance

Total reserved at peak: ~70–110 GB. Headroom: ~15–50 GB.

11.7 Snapshot implementation

Snapshots serialize the entire substrate state. Key implementation considerations:

Staging snapshot. During a consolidation boundary (natural pause between sleep phases), the substrate’s mutable state is copied to a staging buffer. This copy is fast (GPU-resident arrays copied within unified memory) and happens on the main thread. The 60 Hz loop is paused briefly, typically under 100 ms.

Background serialization. The staging buffer is then serialized on background CPU cores using a thread pool. This takes seconds to minutes depending on substrate size but does not block the main loop.

Delta encoding. Warm snapshots (tier 2) are deltas against the most recent full snapshot. The delta includes only changed state: synapses whose weights moved, neurons added or removed, chemistry voxels whose concentrations shifted beyond a small threshold. Deltas are 2-3 orders of magnitude smaller than full snapshots.

Compression. All snapshots are compressed with zstd at level 6 (balance of compression ratio and speed). Typical compression factor: 3–5x for substrate state.

Retention policy. See Section 6 of the companion architecture document; retention is managed by a background process that prunes expired tier-2 and tier-3 snapshots according to configured policy.

11.8 Testing strategy

Three layers:

Unit tests. Each mechanism tested in isolation against reference implementations. Izhikevich dynamics tested against published figures. NMDA plateau dynamics tested against Larkum et al. (2009). STDP behavior tested against Bi and Poo (1998). Chemistry diffusion tested against analytical solutions for known initial conditions.

Integration tests. Multi-hour synthetic runs with scripted input streams. Verify that substrate statistics stay in biological regimes (firing rate distributions, ISI statistics, E/I balance, criticality estimates). Verify that snapshots round-trip cleanly. Verify that structural plasticity does not destabilize the network over many sleep cycles.

Longitudinal tests. Multi-day simulations to check for slow drift, resource leaks, or pathological dynamics that only emerge over long timescales. Budgeted as quarterly verification runs, not per-commit.

11.9 Observability

Every mechanism emits structured telemetry:

- Per-tick timing breakdowns (phase latencies)
- Substrate health metrics (firing rates, criticality, weight norms)
- Plasticity metrics (LTP/LTD rates, eligibility pressure)
- Structural metrics (neurogenesis/apoptosis rates, synapse counts over time)
- Chemistry metrics (per-chemical mean, variance, spatial range)
- Interoception metrics (body variables over time)

Metrics are aggregated in-process and exposed via a Prometheus endpoint for scraping. A companion dashboard visualizes the substrate state for operators.

11.10 Profiling discipline

Python substrates are easy to prototype and easy to slowly degrade. A disciplined profiling practice is essential:

- `py-spy` for sampling profiles, caught regressions in release candidate testing
- `line_profiler` for targeted investigation of specific hot paths
- CuPy and PyTorch profilers for GPU kernel analysis
- Custom per-tick phase timers in the substrate itself

We recommend: every release candidate runs a standard workload profile; any path that grows to consume >5% of tick time is investigated; paths consuming >15% are targeted for optimization.

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14. Expected Performance

This section presents expected performance characteristics. These are projections, not measurements. They are derived from three sources: (1) component-level benchmarks from the literature for similar mechanisms; (2) scaling arguments based on the substrate's computational structure; (3) extrapolation from related systems running on similar hardware.

We mark all numbers as "expected" throughout and provide our reasoning for each estimate.

12.1 Scale targets

On target hardware (16-core CPU, 40-CU iGPU with unified memory, 128 GB RAM, ~50 TOPS NPU):

Configuration	Expected neurons	Expected synapses	60 Hz feasible?
Small (development)	50,000	5 million	Yes, comfortable
Medium (production)	150,000	15 million	Yes, with headroom
Large (research)	250,000	25 million	Yes, tight
Maximum	500,000	50 million	Yes, at edge of envelope

These estimates are derived from:

- Per-EMCN integration cost (~250 FLOPs per sub-step, Section 4.9)
- Synapse propagation cost (dependent on spike rate and connectivity, Section 5.7)
- Chemistry diffusion cost (~2.4 GFLOPs/sec, Section 6.8)
- Available iGPU throughput (~2–4 TFLOPs/sec sustained on this class of iGPU for FP32)
- Available memory bandwidth (~256 GB/s on Strix Halo)

For 250,000 neurons: neuron integration needs 62 GFLOPs/sec, synapse propagation ~1 GFLOPs/sec, chemistry diffusion ~2.4 GFLOPs/sec, plasticity updates ~5 GFLOPs/sec. Total ~70 GFLOPs/sec of substrate compute, or roughly 2–3% of iGPU peak throughput. The tight budget is memory bandwidth and kernel launch overhead, not raw compute.

Our expected scale ceiling is 500,000 neurons. Beyond this, per-neuron state approaches the memory bandwidth limit for 60 Hz operation.

12.2 Frame budget allocation

At 250,000 neurons in steady state (wake):

Phase	Expected cost	Notes
Event queue drain	0.5 ms	Python-side, cheap
Neuron integration (16 sub-steps @ 1 kHz)	3 ms	iGPU, dominant cost
Synapse propagation	2 ms	iGPU, sparse and event-driven
Chemistry diffusion (amortized: every 6th frame)	0.5 ms avg	iGPU, cheap per frame amortized
Plasticity trace updates	1 ms	iGPU
Three-factor weight updates (10 Hz, amortized)	0.3 ms avg	iGPU
Glial + homeostatic (0.1 Hz, amortized)	negligible	
Interoception step	0.2 ms	CPU, trivial
PAD readout, emotion classification	0.2 ms	CPU, trivial
Telemetry and metrics	0.3 ms	CPU
Python-side frame work	3–5 ms	Expression generation, VTS bridge, I/O
Total expected	**~10–12 ms**	Headroom: 4–6 ms for jitter

These numbers are projected from component benchmarks. We expect real measurements to land within ±30% of these estimates. If the estimates prove optimistic, adaptive quality mechanisms (reducing neuron count, increasing chemistry tick period, reducing perception refresh rate) provide graceful degradation.

12.3 Sleep phase performance

Sleep phases have different performance characteristics:

- **NREM**: external input suppressed, so perception layer load drops. Substrate is active with replay events and consolidation, but the main 60 Hz external contract continues. Expected per-frame cost slightly lower than wake due to reduced perception work.

- **REM**: similar to NREM for frame cost. Dream narration via LLM runs as background work at low frequency.
- **Structural plasticity during sleep transitions**: batched operation, runs once per sleep cycle during the NREM-REM boundary. Expected duration: 1–5 seconds, during which the main 60 Hz loop is paused or runs at reduced fidelity. External consumers see a brief pause, equivalent to the agent "blinking" between sleep phases.

Sleep cycles at the biological rate (3–5 per ~8-hour sleep period) mean structural plasticity batches run ~3–5 times per sleep period.

12.4 Memory footprint

Expected memory breakdown for a 250,000-neuron substrate:

Component	Expected size	Notes
Neuron state arrays	125 MB	~500 bytes × 250k
Synapse state arrays	1 GB	~48 bytes × 20M
Connectivity (CSR)	300 MB	indices and pointers
Chemistry field	67 MB	128^3 × 8 × 4 bytes
Glial state	2 MB	sparse, ~25k cells
Plasticity traces	400 MB	per-synapse traces
Delay queue buffers	100 MB	ring buffers
Receptor expression	16 MB	8 × 4 × 250k
Telemetry counters	50 MB	rolling histograms
Auxiliary structures	100 MB	indices, caches
Total expected hot state	**~2.2 GB**	

Snapshots (compressed tier-3 daily) expected size: ~500 MB to 1.5 GB. Journal rate: ~10 MB/min during active wake periods.

12.5 Learning rate expectations

At the default base learning rate ($\eta = 0.01$) with conservative three-factor modulator weights, expected observable learning timescales:

- **Within-session adaptation** (individual synapses strengthening during a 30-minute interaction with high dopamine modulation): subtle but measurable in weight statistics
- **Day-over-day consolidation**: clear differences in weight distributions between morning and evening, reflecting the day's experiences
- **Week-scale preference formation**: emerged preferences for specific input patterns (e.g., specific interaction types) measurable via differential substrate activation
- **Month-scale personality evolution**: observable differences in PAD response characteristics and decision patterns

These timescales match biological learning roughly: humans need weeks to form preferences, months to develop new skills, years to change personality. An agent operating at these rates will feel lifelike on those timescales; acceleration to machine-learning-style hour-scale learning would destabilize the substrate.

For research applications requiring faster observation, the base learning rate can be multiplied by 10x (not more, to preserve stability). This gives day-scale personality evolution, useful for validation studies but not recommended for production.

12.6 Long-term stability

Expected behavior over multi-month continuous operation:

- Total synapse count expected to stabilize within ±20% of initial value after a few months of operation, as structural plasticity reaches equilibrium with input statistics
- Total neuron count expected to stabilize similarly, near the hardware-defined carrying capacity
- Weight distribution statistics expected to shift progressively as consolidated patterns accumulate, then approach a dynamic equilibrium
- Chemistry baseline expected to drift slightly with circadian adaptation to observed schedule

Pathological drift (runaway potentiation, mass apoptosis, chemistry saturation) is expected to be caught by the criticality controller, BCM metaplasticity, and weight-norm watchdog mechanisms. If these fail, rollback to a recent snapshot provides recovery.

12.7 Comparison with related systems

We compare expected performance to three related system classes:

General-purpose SNN simulators (NEST, Brian2, NEURON): These can scale to millions of neurons on cluster hardware. On single consumer workstations, NEST with point neurons achieves ~100k neurons in real time; NEURON with full multi-compartment models achieves ~5k neurons in real time. Our expected 250k multi-compartment EMCNs is between these points, reflecting the reduced-multi-compartment design choice.

ML-oriented SNN frameworks (snnTorch, Norse, BindsNET): Optimized for gradient-based training of point-neuron networks. Achieve high throughput during training but typically not real-time during inference (no need). Our architecture is different in kind — continuous learning with no train/deploy distinction — and is not directly comparable on throughput.

Specialized neuromorphic hardware (Loihi 2, SpiNNaker): Orders of magnitude higher neuron counts per watt than GPU-based simulation. Our substrate could eventually target these platforms but currently targets consumer hardware for accessibility.

We expect our substrate to be the best available choice for the specific use case (continuously-learning embodied agent with biological dynamics on consumer hardware); not the best on any individual axis against specialists.

---

15. Limitations and Open Questions

13.1 Unvalidated mechanisms

Several mechanisms in the substrate are proposed but not empirically validated at scale:

- The three-factor plasticity rule with spatial chemistry gating is novel and we do not have empirical confirmation that it produces the credit assignment behavior we expect
- Unbounded structural plasticity with metabolic competition has been explored in smaller models but not demonstrated at our target scale
- The emergence of functional specialization (hippocampal-like region developing hippocampal-like behavior from developmental dynamics) is plausible but not guaranteed

These are the risk areas for the proposal. Empirical implementation will either confirm or require revision.

13.2 Calibration burden

The substrate has hundreds of parameters. Default values are derived from the literature where possible and from reasoned estimates otherwise. Calibrating a full substrate to produce desired dynamics is expected to be a significant effort — weeks to months of tuning.

We mitigate this by: (a) providing reasonable defaults that should produce stable substrates even if not optimal; (b) exposing calibration through a configuration hierarchy rather than requiring code changes; (c) providing telemetry that makes parameter effects visible.

13.3 Embodiment is simulated

The agent's "body" exists only as simulated variables. We do not claim this is equivalent to physical embodiment. The loop we implement captures the functional structure of interoceptive emotion, but the absence of real sensory-motor grounding is a genuine limitation.

Extending to physical robotic embodiment is straightforward at the interface level (the interoception module could accept real sensor inputs) but introduces many additional engineering challenges beyond the scope of this architecture.

13.4 No subjective experience claim

The substrate produces continuous, adaptive, embodied dynamics. We make no claim that these dynamics constitute subjective experience, consciousness, or sentience. Whether any computational system can have experience is an open philosophical and empirical question; we do not take a position.

What we do claim is that the substrate produces behavior that tracks its own internal state in ways that current stateless systems do not, and that this tracking is mechanical rather than metaphorical.

13.5 Scale ceiling

Target hardware imposes a scale ceiling around 500,000 neurons at 60 Hz. This is small compared to biological brains (roughly 100 billion neurons in human cortex). The substrate is not an attempt to model a brain; it is an attempt to provide a specific class of agent with biologically-inspired dynamics at a scale feasible on consumer hardware.

Scaling up requires either better hardware (neuromorphic accelerators, larger GPUs, cluster deployment) or more aggressive approximations. Neither is addressed in the current proposal.

13.6 Safety of unbounded learning

An agent that is always learning is an agent that can always drift. Our safety mechanisms (criticality controller, BCM metaplasticity, glial scaling, weight-norm watchdog) provide four independent defenses, but none has been validated at scale on this architecture.

We expect additional safety mechanisms to emerge during implementation. The research-scale verification of safety under continuous operation is a significant open question.

13.7 Reproducibility of emergence

Several key behaviors are expected to be emergent from the substrate dynamics (functional specialization, stable preferences, identity persistence under structural turnover). Emergence is notoriously difficult to guarantee from specification. Two substrates with identical configurations but different random seeds will diverge; whether they both produce the desired emergent behaviors is empirical.

This is not unique to our architecture — it is a characteristic of any system with significant stochastic or activity-dependent components — but it is a real limitation on the specifiability of the agent's behavior.

13.8 Open questions we expect to investigate

- Empirical firing rate statistics under steady-state wake conditions
- Measured criticality (branching ratio σ) stability over multi-day runs

- Realized learning rates across multiple modulator configurations
- Emergence of functional specialization from developmental dynamics
- Catastrophic forgetting resistance under distribution shifts
- Memory capacity (how many distinct patterns can be simultaneously learned?)
- Recovery time constants after perturbation (how quickly does mood normalize?)
- Subjective assessment of lifelikeness from human observers (pilot study planned)
- Transferability of substrate snapshots across hardware generations
- Scaling behavior on different hardware classes

Each of these is a legitimate research program. The architecture is intended to enable them, not to answer them in advance.

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16. Conclusion

We have presented a detailed architecture for a continuously-learning embodied neural substrate targeted at synthetic affective agents. The architecture centers on the Embodied Multi-Compartment Neuron (EMCN) — a three-compartment spiking neuron with Izhikevich soma dynamics, NMDA-dependent apical plateaus, a metabolic budget, stochastic channel gating, and coupling to a 3D chemistry field via receptor expression. Neurons communicate through conductance-based stochastic synapses with per-synapse delays. A diffusing 3D chemistry substrate replaces scalar modulator variables, allowing spatial credit assignment and making PAD emotional state a readout rather than a primary variable. Plasticity combines STDP with chemistry-gated three-factor learning, BCM metaplasticity, and sleep-gated structural plasticity that supports unbounded neurogenesis and apoptosis bounded only by the hardware envelope. An interoceptive body-state loop closes the embodiment circle, making emotional dynamics emergent from substrate-body-chemistry interactions rather than externally assigned.

The architecture targets a single consumer workstation with heterogeneous compute (CPU, iGPU, NPU) and unified memory, and is expected to support 150,000 to 500,000 EMCN neurons with 15–50 million synapses at a 60 Hz external contract. Implementation is in Python with tiered acceleration (NumPy, Numba, CuPy, PyTorch-ROCm) and uses arena-based slotmaps with generational indexing to support the unbounded structural plasticity.

We do not claim experimental validation. We claim a coherent design that integrates known mechanisms in a novel combination aimed at enabling behaviors (continuous learning, substrate-level memory of individuals, emotional persistence, sleep-dependent consolidation, identity under structural change) that current AI architectures cannot support in any mechanical sense. Empirical validation is the next step.

We release this architecture specification openly for the benefit of developers building long-running AI agents. We welcome collaboration on implementation, validation, and extension.

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Appendix A: Parameter Reference Table

This appendix provides a consolidated table of recommended starting parameter values for the substrate. Values are derived from the literature where possible and from reasoned estimates otherwise. These are starting points for implementation, to be refined through empirical tuning.

A.1 EMCN neuron parameters (regular spiking pyramidal baseline)

Parameter	Symbol	Value	Units	Source
Izhikevich a	a	0.02	—	Izhikevich 2003
Izhikevich b	b	0.2	—	Izhikevich 2003
Izhikevich c	c	-65	mV	Izhikevich 2003
Izhikevich d	d	8	—	Izhikevich 2003
Spike threshold	v_{peak}	30	mV	Izhikevich 2003
Rest potential	v_{rest}	-70	mV	cortical baseline
Apical time constant	τ_a	30	ms	Larkum et al. 2009
Basal time constant	τ_b	15	ms	cortical baseline
Soma-apical coupling	$g_{\text{ax,a}}$	0.5	—	tuned
Soma-basal coupling	$g_{\text{ax,b}}$	0.8	—	tuned
NMDA plateau threshold (V)	V_{plat}	-40	mV	Larkum et al. 2009
NMDA plateau threshold (Ca)	Ca_{plat}	0.5	µM	reasoned
Calcium decay	τ_{Ca}	100	ms	cortical baseline
Baseline ATP rate	k_{base}	0.001	/s	reasoned
Per-spike ATP cost	k_{spike}	0.01	—	Attwell & Laughlin 2001 scaled
Glial ATP supply rate	k_{supply}	0.05	/s	reasoned, tuned to baseline

A.2 Synapse parameters (AMPA-like excitatory)

Parameter	Symbol	Value	Units	Source
Maximum conductance	g_{max}	0.5	nS	cortical baseline
Rise time	τ_{rise}	0.5	ms	cortical baseline
Decay time	τ_{decay}	5	ms	cortical baseline

Parameter	Symbol	Value	Units	Source
Reversal potential	E_{rev}	0	mV	AMPA
Baseline release probability	P_{release}	0.4	—	cortical baseline
Vesicle pool maximum	N_{max}	5	—	cortical baseline
Vesicle replenishment time	$\tau_{\text{replenish}}$	500	ms	cortical baseline
Baseline delay	d_{baseline}	1	ms	cortical baseline
Conduction velocity	v_{cond}	0.5	m/s	local unmyelinated

A.3 Chemistry parameters

See Section 6.3 main text. All chemical-specific parameters listed there.

A.4 Plasticity parameters

Parameter	Symbol	Value	Source
LTP amplitude	A_{pre}	0.005	Bi & Poo 1998 scaled
LTD amplitude	A_{post}	-0.00525	Bi & Poo 1998 scaled
STDP pre trace time constant	τ_{pre}	20 ms	Bi & Poo 1998
STDP post trace time constant	τ_{post}	20 ms	Bi & Poo 1998
Eligibility trace time constant	τ_e	1000 ms	Gerstner et al. 2018
Base learning rate	η	0.01	tuned
Dopamine plasticity weight	w_D^{plast}	1.0	reasoned
Acetylcholine plasticity weight	w_A^{plast}	0.6	reasoned
Cortisol plasticity weight (negative)	w_C^{plast}	-0.5	Kim & Diamond 2002
BCM time constant	τ_{BCM}	60 s	Bienenstock et al. 1982

A.5 Criticality target

Parameter	Symbol	Value
Branching ratio target	σ_{target}	1.0
Deadband	ϵ	0.05
Controller time constant	τ_{crit}	60 s

A.6 Structural plasticity rates

Operation	Per-cycle rate
Neurogenesis	0.1% – 1% of population
Apoptosis	0.1% – 1% of population
Synaptogenesis	1% – 5% of synapse count
Synaptic pruning	1% – 5% of synapse count

Appendix B: Glossary of Terms

Apoptosis — programmed neuron death in the substrate, triggered by metabolic insufficiency, consolidation redundancy, critical-period closure, microglial elimination, or resource competition.

ATP budget — per-neuron scalar representing metabolic availability; modulates threshold and plasticity at low levels.

Arena-based slotmap — dynamic allocation pattern with chunked state arrays, free list of reclaimed slots, and per-slot generation counter; supports unbounded add and remove.

BCM threshold — sliding activity threshold that prevents runaway potentiation; higher for chronically active neurons.

Chemistry field — 3D voxel grid of diffusing chemical concentrations; replaces scalar modulator variables.

Compartment — a neuron subdivision with its own voltage dynamics. EMCN has three: soma, apical dendrite, basal dendrite.

Criticality — branching ratio σ of the network; target σ ≈ 1 for healthy dynamics.

EMCN — Embodied Multi-Compartment Neuron; the unit neuron of the substrate.

Eligibility trace — per-synapse scalar integrating recent STDP events, converted to weight change when chemistry gates plasticity.

Generational ID — ID pair (slot, generation) allowing slot reuse without aliasing; old IDs become stale when their slot's generation increments.

Interoception — simulated body-state signals that feed into chemistry; source of embodied emotion.

NMDA plateau — prolonged dendritic depolarization from NMDA receptors when voltage and calcium co-occur; coincidence detection mechanism.

PAD — pleasure-arousal-dominance, the three-dimensional emotional state representation; computed as a readout from chemistry.

Receptor expression vector — per-neuron vector specifying sensitivity to each chemical; the mechanism by which chemistry modulates neuron parameters.

Substrate — the neural system (neurons, synapses, chemistry, glia, plasticity, and associated mechanisms); the novel contribution of this architecture.

Three-factor plasticity — synaptic update rule where the third factor is local chemistry at the synapse; enables spatially heterogeneous credit assignment.

End of document.